PARIKSHA365 — STUDY NOTES

# Science & Technology — Common Book

SSC, RRB, Banks (GA), State PSC — ISRO, DRDO, Computers, Defence Tech

SSC RRB BANKS STATE PSC

v 2026-04

**The promise.** Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

# EXAMINER BLUEPRINT — What Gets Tested Most

**Science & Technology is one of the most scoring sections.**

- Questions are almost entirely **factual**.
- No derivation, no calculation.
- Just "Which mission did what" or "Which organisation developed X."
- Knowing the right list gets you full marks.

EXAM ALERT

**Top 8 Sci-Tech exam hotspots**

1. **ISRO missions** — Chandrayaan series, Mangalyaan, Aditya-L1, Gaganyaan, SpaDeX. Know each mission's objective, launch vehicle, and year. Appears in every SSC, RRB, and Banking exam.
2. **DRDO missiles** — Agni series, Prithvi series, BrahMos, Akash, Tejas. Know range, type (surface-to-surface, supersonic cruise), and developer.
3. **Nobel Prize winners** — last 3–5 years across Physics, Chemistry, Medicine/Physiology. Examiners ask year + field + discovery.
4. **Computer fundamentals** — generations of computers, OSI model (7 layers), storage units (1 byte = 8 bits), programming generations.
5. **Biotechnology** — PCR, CRISPR, cloning, vaccines (live attenuated vs killed vs mRNA). These appear in SSC and UPSC papers.
6. **Nuclear technology** — fission vs fusion, types of reactors (PWR, BWR, PHWR), India's 3-stage nuclear programme, BARC vs ISRO vs DRDO roles.
7. **Defence platforms** — INS Vikrant (aircraft carrier), Tejas, AMCA, Arjun tank, Arihant-class submarines.
8. **Space organisations** — NASA (USA), ESA (Europe), Roscosmos (Russia), CNSA (China), JAXA (Japan). Know 2–3 key missions each.

KEY FACT

**Study priority for time-poor students**

1. ISRO missions — all with years and objectives
2. DRDO missiles — Agni I through V ranges, BrahMos specs
3. Nobel Prize winners — last 3 years
4. Computer fundamentals — OSI model, storage, generations
5. Biotechnology basics — vaccines, CRISPR, DNA vs RNA
6. India's nuclear programme — 3 stages + key reactors
7. Defence platforms — aircraft carrier, submarines, fighter jets
8. Space milestones (global) — first human in space, Moon landing, Mars missions

## What's New This Year (Sci-Tech)

> *ISRO/DRDO missions, Nobel laureates, AI/tech companies' product versions, Indian-tech milestones change frequently. Re-check before any exam later than Q3 2026. The rest of this book (general physics/chemistry/biology fundamentals, computer-architecture basics, OSI 7 layers, programming-language histories, MS Office shortcuts, internet milestone history pre-2024) is* ***static and stable****.*

| Item | Current value (verify before your exam) |
| --- | --- |
| **Nobel 2025 — Physics** | John Clarke + Michel H. Devoret + John M. Martinis (macroscopic quantum tunnelling) |
| **Nobel 2025 — Chemistry** | Susumu Kitagawa + Richard Robson + Omar M. Yaghi (Metal-Organic Frameworks) |
| **Nobel 2025 — Medicine** | Mary Brunkow + Frederick Ramsdell + Shimon Sakaguchi (regulatory T cells) |
| **ISRO recent missions** | Chandrayaan-3 (23 Aug 2023 lunar landing); Aditya-L1 (Sep 2023, L1 reached Jan 2024); SpaDeX (30 Dec 2024, docking 16 Jan 2025); NISAR (30 Jul 2025 GSLV-F16) |
| **Gaganyaan** | Uncrewed G1 = H2 2026 (with Vyommitra robot); first crewed mission = H1 2027 |
| **DRDO recent** | Mission Divyastra (Agni-V MIRV, Mar 2024); Agni-V test Aug 2025; Scramjet long-duration test Jan 2026 |
| **Defence platforms** | INS Vikrant (commissioned 2 Sep 2022); INS Arighat (Aug 2024); HAL Tejas Mk-1A (delivery 2024) |
| **CBDC** | Digital Rupee (e₹): Wholesale pilot Nov 2022; Retail pilot Dec 2022; live in 9 cities |
| **5G** | India 2nd-largest 5G user base; 394 million users end-2025; 1 billion broadband subscribers Nov 2025 |
| **CRISPR therapy** | Casgevy (first CRISPR therapy approved by FDA Dec 2023) — sickle cell + β-thalassemia |

## How to use this book

**Sci-Tech questions are almost entirely about four things:**

- **Names** — ISRO, DRDO, BARC, BrahMos, INS Vikrant.
- **Years** — when launched / commissioned.
- **Organisations** — who developed it.
- **One-line description** — range, payload, purpose.

**No calculation required.** This makes Sci-Tech the easiest scoring section — IF you commit the right tables to memory.

**Why each chapter starts with a story:**

- Understanding **why** a system was built makes the specific facts (range, year, mechanism) **stick far better** than raw memorisation.
- Read the story → drill the table at the end.

**The "Exam hooks" section is your most important page in every chapter:**

- It is the **compressed list** of exactly what examiners have asked.
- After reading, **close the book**, answer those points from memory.
- If you can → move on.
- If you cannot → re-read only the section you struggled with, and try again.

> ***The truth about Sci-Tech:*** *Students who can recall "BrahMos: 290 km range, Mach 2.8, surface-surface-air-sea, Indo-Russian joint venture" in 5 seconds beat students who "understood" it perfectly but can't retrieve it under pressure. This book is built to get you to that 5-second recall speed.*

## How to read this book — your real study load

**Total pages: 58.** Don't let the page count scare you.

### Mandatory reading — pages 1 to 29 (50% of the book)

That's the full syllabus. Master those 29 pages and you've covered every concept an examiner can fairly ask.

### Bonus material — pages 30 to 58 (50% of the book)

Reference tables, compilations, and self-test material — useful in your last week of revision OR if you're aiming for a top rank.

### Two study tracks

- **Just want to pass?** Read pages 1–29. Skim the appendices once before the exam.
- **Want to top your batch?** Read the appendices too — they're how 70% becomes 90%.

## Index — Table of Contents (clickable)

# PART A — SPACE TECHNOLOGY: INDIA'S JOURNEY FROM THUMBA TO THE MOON

## Chapter A1 — The Story of ISRO: From Coconut Groves to Cosmos

### Background

**From bicycles to the moon — ISRO's arc.**

- **1962, Thumba, Kerala** — a fishing village; a Jesuit church converted into a rocket workshop.
- Parts carted on **bicycles**.
- First Indian rocket — a Nike-Apache — launched **21 November 1963**.
- **63 years later** — Chandrayaan-3 lands near the **lunar south pole** (23 Aug 2023).
- India becomes the **first country** ever to do so.

### Why India needs a space programme

**Vikram Sarabhai's vision (1963):**

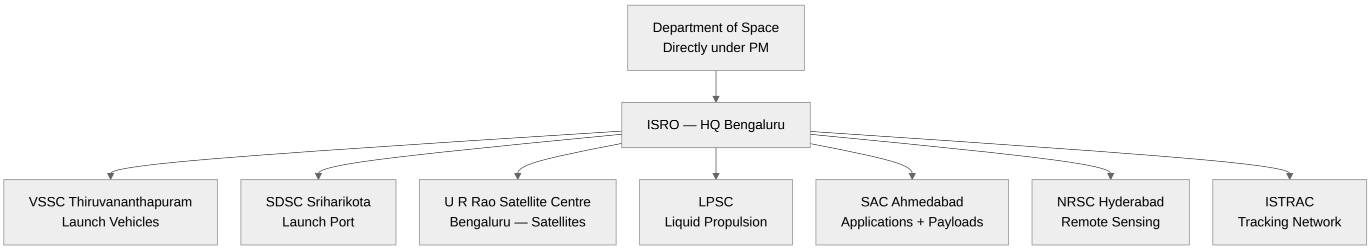
> *"There is no ambiguity of purpose. We must be **second to none** in the application of advanced technologies to the real problems of man and society."*

**Translation** — India's space programme exists for **development**, not prestige:

- Weather forecasting
- Communication (TV, telephony)
- Education (EDUSAT)
- Resource mapping (RESOURCESAT)

- Disaster response

## 🔭 Organisational Tree




## 🚀 Launch Vehicles — The PSLV → GSLV → LVM3 → SSLV Ladder

| Vehicle | Full form | Payload LEO | Signature role |
|---|---|---|---|
| **SLV-3** (1980) | Satellite Launch Vehicle | 40 kg | First Indian launcher (Rohini D1) |
| **ASLV** (1987-94) | Augmented SLV | 150 kg | Learning platform |
| **PSLV** (1993-) | Polar Satellite Launch Vehicle | 1750 kg (SSO) | ISRO's warhorse — 104 sats in one go (2017) |
| **GSLV Mk-II** (2001-) | Geosync SLV | 2.5 t (GTO) | Cryogenic upper stage (indigenous from 2014) |
| **LVM3 (GSLV Mk-III)** | Launch Vehicle Mark 3 | 4 t (GTO) / 8 t (LEO) | Chandrayaan-2, Chandrayaan-3, Gaganyaan |
| **SSLV** (2022-) | Small Satellite LV | 500 kg (LEO) | Cheap, on-demand launches |

### Memory aid: "The Launchpad Walk"

Picture yourself walking from the gate of **Satish Dhawan Space Centre, Sriharikota** toward the sea:

1. **Gate** → a **small** rocket waves at you = **SSLV** (small).
2. **Assembly building** → the **warhorse PSLV** is being stacked (**P**olar orbits, **S**un-sync).
3. **Next pad** → **GSLV** stands taller with its cryogenic nose (**G**eostationary, cold fuel).
4. **Main launch tower** → **LVM3**, the giant that carried Chandrayaan-3.
5. **Beach** → out in the ocean, a future **Gaganyaan** crewed capsule splashes down.

Walk this path in your head three times — you will never forget the launcher hierarchy.

## 🛰️ Mission Chronicle — the milestones every exam asks

| Year | Mission | What mattered |
|---|---|---|
| 1975 | **Aryabhata** | First Indian satellite (launched by USSR) |
| 1980 | **Rohini-1** | First satellite launched by an Indian rocket (SLV-3) |
| 1983 | **INSAT-1B** | Beginning of INSAT comm series |
| 1988 | **IRS-1A** | Earth Observation series begins |

| Year | Mission | What mattered |
|---|---|---|
| 2008 | **Chandrayaan-1** | Confirmed water molecules on Moon |
| 2013 | **Mangalyaan / MOM** | India = **first Asian nation to reach Mars**, first country to do so on maiden attempt |
| 2014 | **GSLV-D5** | First successful indigenous cryogenic stage |
| 2017 | **PSLV-C37** | 104 satellites in one launch (world record) |
| 2019 | **Chandrayaan-2** | Orbiter success; lander (Vikram) crashed |
| 2020 | **RISAT-2BR1** | Radar-imaging series |
| 2023 | **Chandrayaan-3** | Soft-landing near lunar south pole (23 Aug) — India = 4th country on Moon, 1st near south pole |
| 2023 | **Aditya-L1** | India's first solar observatory (Lagrange-1 point) |
| 2024 | **XPoSat** | X-ray polarimetry satellite (New Year's Day) |
| 2025 | **SpaDeX** | Space Docking Experiment — docking capability demo |

### Chandrayaan-3 — Mission Details

- **Launch:** 14 July 2023, LVM3-M4, Sriharikota.
- **Arrival:** Lunar orbit 5 Aug; landing 23 Aug 2023, 6:04 PM IST.
- **Landing site** named **Shiv Shakti Point**; Chandrayaan-2 crash site named **Tiranga Point**.
- **National Space Day:** declared for **August 23** every year.
- **Lander:** Vikram. **Rover:** Pragyan (6-wheeled, 26 kg).
- **Firsts:** First near lunar south pole; 4th country to soft-land (after USA, USSR, China).

### 🔆 Aditya-L1

- Launched **2 Sep 2023** on PSLV-C57.
- Placed at **Lagrange Point 1 (L1)**, ~1.5 million km from Earth toward the Sun (continuous Sun view, no eclipses).
- **Seven payloads** — VELC (flagship), SUIT, ASPEX, PAPA, SoLEXS, HEL1OS, Magnetometer.

### 🧭 Navigation, Communication, Earth Observation Constellations

- **NavIC** (IRNSS): 7 satellites, regional navigation for India + 1500 km around.
- **GAGAN:** GPS Aided Geo-Augmented Navigation (for civil aviation, joint with AAI).
- **INSAT + GSAT:** communication & broadcasting.
- **IRS / Cartosat / Resourcesat / Oceansat / RISAT:** Earth observation.

### 🚀 Gaganyaan Programme

- India's first human spaceflight mission. Target: **2026**.
- Crew: **3 astronauts**, **3 days**, **LEO (~400 km)**.
- **Four astronaut-designates** (announced Feb 2024): Group Captains Prashanth Nair, Ajit Krishnan, Angad Pratappa, Shubhanshu Shukla.

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- **Abort tests** (TV-D1) already conducted.
- Launcher: **Human-rated LVM3 (HLVM3)**.

### 🌌 Future Missions Roadmap

- **Mangalyaan-2** (MOM-2) — next Mars mission.
- **Shukrayaan-1** — Venus orbiter (~2028).
- **Chandrayaan-4** — lunar sample return.
- **Bhartiya Antariksh Station (BAS)** — Indian space station by 2035.
- **Crewed lunar landing** by 2040 (announced by PM, Oct 2023).

### Exam pointers

- "First country to land near lunar south pole?" → **India (Chandrayaan-3, Aug 23 2023).**
- "India's first solar mission?" → **Aditya-L1, at L1 Lagrange point.**
- "PSLV world record for satellites in one launch?" → **104 (C37, 2017).**
- "India's first Mars mission?" → **Mangalyaan (MOM), 2013, success on maiden attempt.**
- "Cryogenic engine first success?" → **GSLV-D5, 2014.**

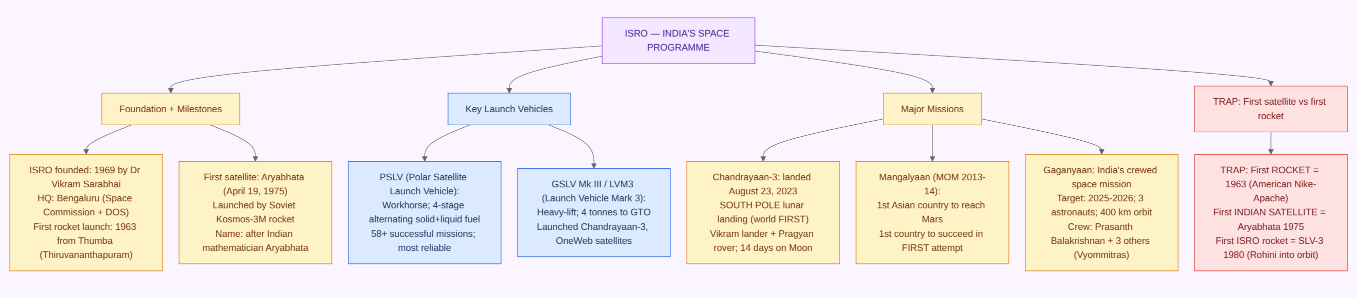
### In simple words

Explain to a class 10 student: **PSLV** is India's workhorse rocket — it launches small satellites into a special orbit that lets them see every spot on Earth as it spins (polar). **GSLV** uses a super-cold engine (cryogenic) to throw bigger satellites far up (36,000 km) where they hover over one spot (geostationary). **LVM3** is India's biggest rocket — it took our lander to the Moon.

### Practice

1. Name ISRO's four active launch vehicles and one signature mission of each.
2. What is Lagrange Point 1, and why did India choose it for Aditya-L1?
3. Which two countries preceded India in soft-landing on the Moon, and which continent was India first in for Mars?

**◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW**




## Chapter A2 — Private Space & Global Context

### 🏭 Private Space Sector Reforms

- **2020:** IN-SPACe (Indian National Space Promotion & Authorization Centre) created under DoS; nodal agency for private space activity.
- **NSIL** (New Space India Limited): commercial arm of ISRO (since 2019).
- **Space Policy 2023:** allows private companies to build rockets, satellites, ground systems; FDI up to **100% automatic** in components, **74%** in satellites, **49%** in launch vehicles.
- **Private startups:** Skyroot (Vikram-S, 1st private rocket, 2022), Agnikul (Agnibaan SOrTeD, 2024 — 1st private sub-orbital with 3D-printed engine), Pixxel, Dhruva, Bellatrix.

### Global Space Context

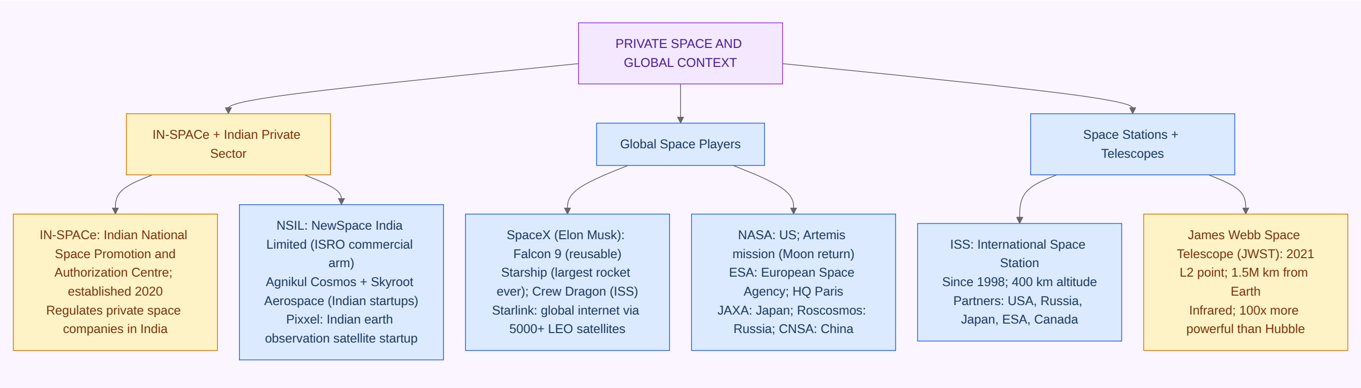
- **SpaceX (USA)** — Falcon 9 reusable, Starship.
- **NASA's Artemis** — return humans to Moon (Artemis III target ~2026).
- **China** — Tiangong station, Chang'e sample returns, Tianwen-1 Mars.
- **ESA** — Ariane 6, JUICE to Jupiter.
- **Russia** — Luna-25 crashed on Moon (Aug 2023, two days before Chandrayaan-3 landed).

### Exam pointers

- "India's 1st private rocket?" → **Vikram-S (Skyroot, 18 Nov 2022, sub-orbital)**.
- "Nodal body for private space?" → **IN-SPACe**.
- "Commercial arm of ISRO?" → **NSIL**.

# PART B — DEFENCE TECHNOLOGY: GUARDING THE NATION

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW




## Chapter B1 — DRDO and the Missile Story

### Background

**1983, Hyderabad.** Dr A.P.J. Abdul Kalam, then DRDO Director, pitches the **Integrated Guided Missile Development Programme (IGMDP)** to PM Indira Gandhi. Five missiles, one indigenous project, decades of sanctions. The programme officially concludes in 2008 — having delivered every promised system and birthing India's strategic shield.

### IGMDP — the original five

| Missile | Type | Range |
|---|---|---|
| **Prithvi** | Surface-to-Surface (short range) | 150-350 km |
| **Agni** | Surface-to-Surface (medium → ICBM) | 700 → 5000+ km |
| **Trishul** | Short-range SAM | 9 km |
| **Akash** | Medium-range SAM | 25-45 km |
| **Nag** | Anti-tank | 4-20 km |

Mnemonic: **PATAN** = **P**rithvi, **A**gni, **T**rishul, **A**kash, **N**ag.

### 🚀 Agni Series Ladder

| Variant | Range | Notes |
|---|---|---|
| Agni-I | 700 km | Single-stage, solid fuel |
| Agni-II | 2000 km | Two-stage |
| Agni-III | 3500 km | Reaches deep into China |
| Agni-IV | 4000 km | Road-mobile |
| **Agni-V** | 5000+ km | ICBM class, canister-launched |

| Variant | Range | Notes |
|---|---|---|
| **Agni-P (Prime)** | 1000-2000 km | Next-gen replacing Agni-I/II |
| **Agni-V MIRV** (2024) | 5000+ km | "Mission Divyastra" — first Indian MIRV test |

### 🌊 BrahMos — The Speed Demon

- Indo-Russian joint venture (India + Russia; named after **Brah**maputra + **Mos**kva).
- **Supersonic cruise missile**, Mach 2.8-3.0.
- Fired from land, ship, submarine, and air (Su-30MKI).
- **BrahMos-II / Hypersonic version** in development (Mach 7+).
- **First export** (Jan 2022): to **Philippines** ($375 mn deal).

### 🛡️ Air-defence & BMD

- **Akash-NG** — next-gen Akash.
- **QRSAM** — Quick-Reaction SAM.
- **Barak-8** — long-range naval SAM (Indo-Israeli).
- **BMD (Ballistic Missile Defence):** two-tier — **PAD/PDV (exoatmospheric)** + **AAD (endoatmospheric)**.
- **Mission Shakti** (27 Mar 2019): ASAT test — India became **4th country** with ASAT capability.

### Memory aid: "The Missile Shed"

Visualise walking into a large shed at DRDO's Chandipur range. Five doors labelled **P-A-T-A-N**:

1. **Prithvi** door — short-range green crate.
2. **Agni** door — ladder inside with rungs I→V→P→MIRV.
3. **Trishul** door — three prongs (SAM, short range).
4. **Akash** door — sky-blue crate (SAM, medium).
5. **Nag** door — anti-tank teeth.

Then an annex: **BrahMos** in a red-white-blue Russo-Indian crate, **Shakti** (ASAT) banner on the ceiling.

### 🛩️ Aeronautics — LCA, AMCA, Drones

- **LCA Tejas (HAL)** — indigenous 4.5-gen fighter. Mk-1A under induction. Mk-2 planned.
- **AMCA** — Advanced Medium Combat Aircraft (5th-gen stealth, in development).
- **HAL Prachand** (formerly LCH) — Light Combat Helicopter, high-altitude specialist.
- **Dhruv & Rudra** — Advanced Light Helicopter family.
- **UAVs:** Nishant, Lakshya (target drone), **Rustom-II/TAPAS** (MALE), **Archer** (armed).

### ⚓ Naval Programme

- **INS Vikrant** (commissioned Sep 2022) — India's **first indigenous aircraft carrier**.
- **INS Arihant-class** — SSBN (nuclear ballistic subs); Arihant (2016), Arighaat (2024).
- **Project 75 (Scorpene)** — Kalvari-class submarines.
- **Project 75-I, 17A (Nilgiri-class frigates), P-15B Visakhapatnam-class destroyers.**

### 🏹 Army Programme

- **Arjun Mk-1A** — indigenous MBT.
- **K9 Vajra** — 155mm self-propelled howitzer.
- **Dhanush** — indigenised Bofors.
- **Pinaka** — multi-barrel rocket launcher.
- **ATAGS** — Advanced Towed Artillery Gun System (DRDO).

### Exam pointers

- "IGMDP — five original missiles?" → **Prithvi, Agni, Trishul, Akash, Nag (PATAN).**
- "BrahMos named after?" → **Brahmaputra + Moskva.**
- "India's ASAT test?" → **Mission Shakti, 27 Mar 2019.**
- "First indigenous aircraft carrier?" → **INS Vikrant (2022).**
- "Agni-V MIRV mission name?" → **Mission Divyastra (2024).**
- "First BrahMos export customer?" → **Philippines (2022).**

### In simple words

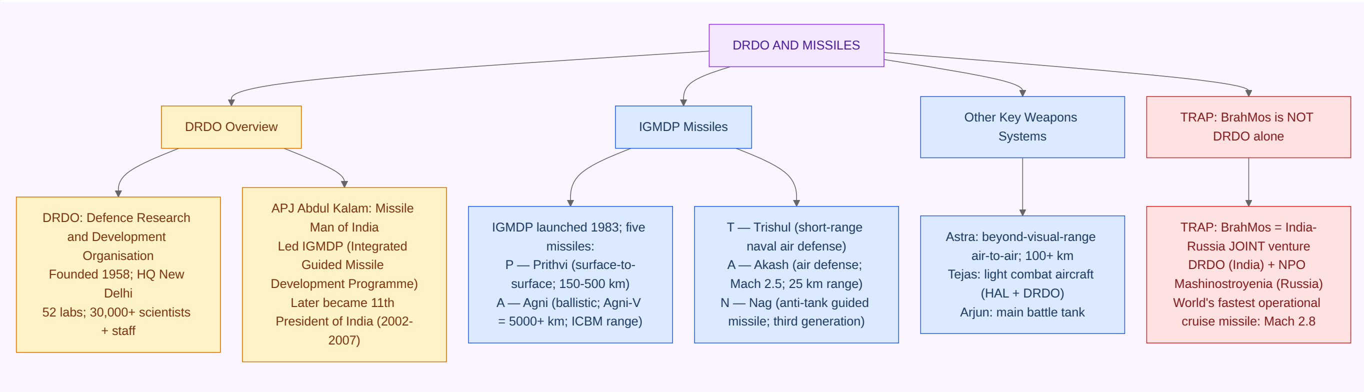
Imagine India has three shields: **Prithvi** for neighbours up close, **Agni** for threats far away, and **Akash/BrahMos** to shoot things *coming at us* out of the sky. Behind the shields stand **Tejas** planes, **Arihant** submarines, **Vikrant** aircraft carriers. Everything was designed here, under sanctions — which is why *indigenisation* is the single word DRDO repeats.

### Practice

1. List PATAN and one salient fact about each.
2. Why is Agni-P considered "next-gen" compared to Agni-I?
3. Compare BrahMos and Tomahawk (speed, cost, range). Why is supersonic an advantage?

# PART C — NUCLEAR PROGRAMME

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW

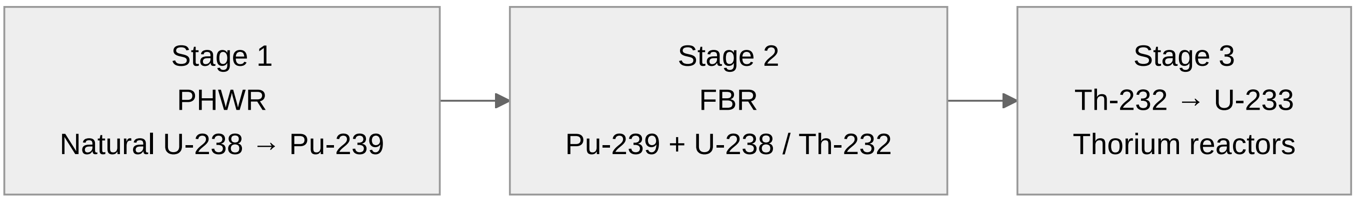



## Chapter C1 — Bhabha's Three-Stage Plan

### Background

**1954, Mumbai. Homi Bhabha** founds the Department of Atomic Energy (DAE). His vision: India, blessed with little uranium but mountains of thorium (~25% of world reserves), will build a **three-stage programme** that turns thorium into fuel by mid-century.

### The Three Stages




- **Stage 1** — Pressurised Heavy Water Reactors (PHWR) use natural uranium, moderated by heavy water ($D_2O$). 18 reactors of this type operate in India.
- **Stage 2** — Fast Breeder Reactors (FBR) use plutonium from stage 1, "breeding" more fuel. **PFBR at Kalpakkam** reached criticality in 2024.
- **Stage 3** — Thorium-based Advanced Heavy Water Reactors (AHWR) — still under design.

### ☢️ Key Nuclear Tests

- **Pokhran-I** (18 May 1974) — Operation **Smiling Buddha**, "peaceful nuclear explosion".
- **Pokhran-II** (11-13 May 1998) — Operation **Shakti**, 5 tests; India declared a nuclear-weapon state. **May 11 → National Technology Day.**

### 🏭 Organisations under DAE

- **BARC** (Bhabha Atomic Research Centre) — Mumbai.
- **IGCAR** (Indira Gandhi Centre for Atomic Research) — Kalpakkam.
- **NPCIL** (Nuclear Power Corporation of India Ltd) — operates reactors.
- **AERB** (Atomic Energy Regulatory Board) — safety regulator.

- **BHAVINI** — FBR operator.

### 🔋 Nuclear Power Capacity (Apr 2026)

- **~7.5 GW** installed (target: 22.4 GW by 2031).
- Major sites: Tarapur (first, 1969), Kaiga, Kakrapar, Kudankulam (Russian VVER), Rawatbhata, Narora, Kalpakkam.
- **Nuclear Liability Act 2010** — controversial "right of recourse" to suppliers.
- **India-US 123 Agreement** (2008) — ended nuclear isolation; enabled NSG waiver.

### Exam pointers

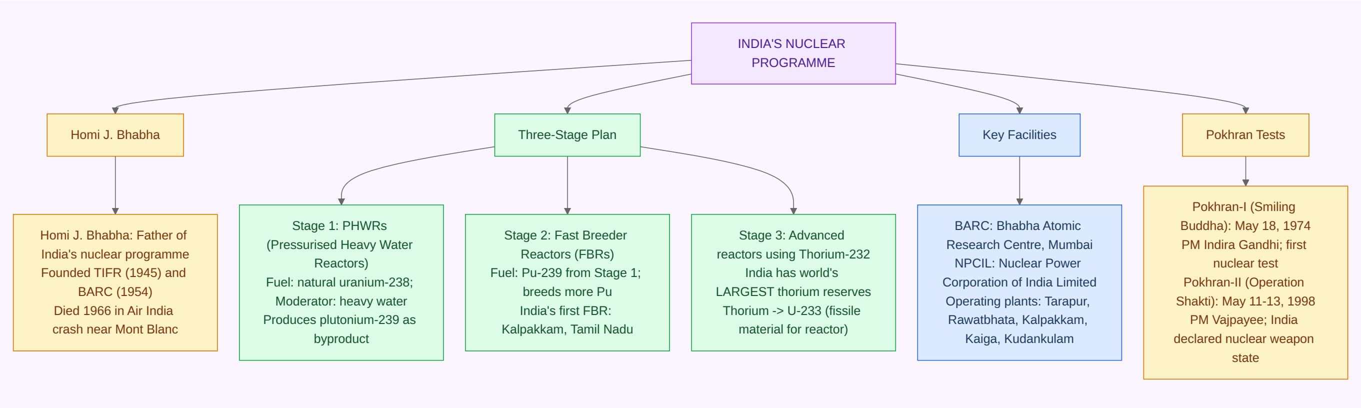
- "Three stages?" → **PHWR → FBR → Thorium.**
- "Pokhran-II?" → **May 11-13, 1998, Operation Shakti → National Technology Day (May 11).**
- "India's thorium share?" → **~25% of world reserves (Kerala beach sands).**
- "Father of Indian atomic programme?" → **Homi Bhabha.**

### Practice

1. Why does India need a thorium cycle?
2. What is the role of heavy water in PHWR?
3. Name three sites and their reactor types.

# PART D — COMPUTING, AI, AND DIGITAL INDIA

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW

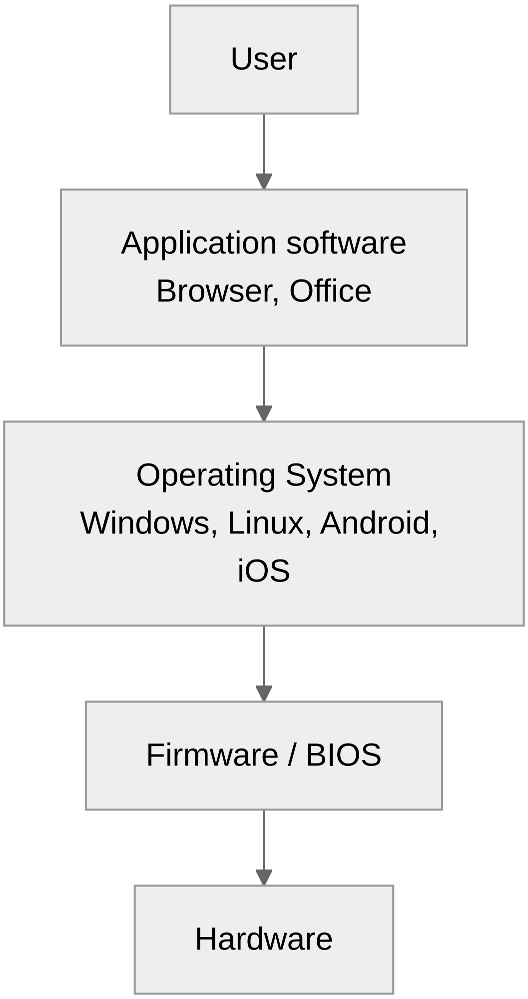



## Chapter D1 — Computer Fundamentals

### 🔤 Basic Vocabulary (non-negotiable for SSC/Banking)

- **Hardware** (physical) vs **software** (programs).
- **CPU** = Central Processing Unit = **brain** (ALU + Control Unit + Registers).
- **RAM** (volatile, fast) vs **ROM** (non-volatile, fixed).
- **Primary storage** (RAM, cache) vs **secondary** (HDD, SSD, USB).
- **Bit** → **Byte** (8 bits) → **KB (1024 B) → MB → GB → TB → PB.**
- **Motherboard** connects everything.
- **GPU** — graphics processor (parallel, heavy in AI now).

## 💾 Software Layers




## Networking

- **LAN → MAN → WAN** (scale ladder).
- **IP v4 (32-bit)** vs **IP v6 (128-bit)**.
- **TCP/IP** — reliable transport.
- **HTTP/HTTPS** — web.
- **DNS** — name → IP.
- **OSI model** — 7 layers (Physical, Data link, Network, Transport, Session, Presentation, Application). Mnemonic: **Please Do Not Throw Sausage Pizza Away**.

## 👾 Malware Zoo

- **Virus** — attaches to files, needs user execution.
- **Worm** — self-replicating over networks.
- **Trojan** — disguised.
- **Ransomware** — encrypts & demands payment (**WannaCry** 2017).
- **Spyware / Adware / Rootkit / Keylogger.**
- **Phishing / Smishing / Vishing** — social engineering.

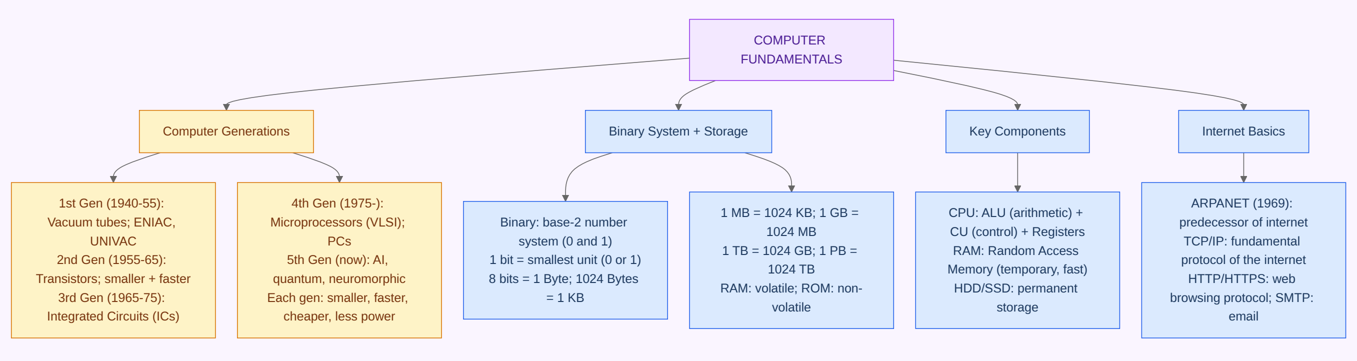
### 🔐 Cryptography Primer

- **Symmetric** (AES) vs **asymmetric** (RSA, ECC) keys.
- **Hash functions** — SHA-256, MD5 (broken).
- **Digital signature** = hash + private-key encryption.
- **TLS/SSL** — secures HTTP → HTTPS.
- **Blockchain** — chained blocks of hashes (Bitcoin 2009 by Satoshi Nakamoto).

### Exam pointers

- "1 KB = ?" → **1024 bytes.** (Some contexts: 1000.)
- "OSI layers count?" → **7.**
- "Father of computers?" → **Charles Babbage (Difference/Analytical Engine).**
- "First computer virus?" → **Creeper (1971).**

**◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW**




## Chapter D2 — Artificial Intelligence & Emerging Tech

### 🤖 AI Taxonomy

- **Narrow AI** (today — ChatGPT, Alexa, image recognition).
- **AGI** (human-level general intelligence — not yet).
- **ASI** (super-intelligent — speculative).

### Key Sub-fields

- **Machine Learning** — learning from data.
- **Supervised** (labelled), **unsupervised** (clustering), **reinforcement** (reward).
- **Deep Learning** — multi-layer neural networks (CNNs for images, RNN/Transformers for sequences).
- **Generative AI** — LLMs (GPT, Claude, Gemini), image models (DALL-E, Midjourney).
- **NLP** — text understanding.
- **Computer Vision** — image/video understanding.
- **Robotics** — physical AI.

### 🇮🇳 India's AI Ecosystem

- **IndiaAI Mission** (Mar 2024): ₹10,371 crore — GPU compute, foundation models, datasets, applications, startups, skilling, safe-and-trusted AI.
- **BHASHINI** — national AI-based language translation platform (22 scheduled languages).
- **AIRAWAT** — AI supercomputer at C-DAC Pune.
- **NITI Aayog's "AI for All" strategy (2018)**.

### Other Emerging Tech

- **Blockchain + Web3** — decentralised ledgers. **CBDC (Digital Rupee)** pilot by RBI (2022).
- **Quantum Computing** — **National Quantum Mission** (2023), ₹6,003 cr, 2023-31; targets 1000-qubit computer.
- **IoT (Internet of Things)** — sensor-networks, smart cities.
- **5G** — rolled out Oct 2022 by PM; 6G Vision Document released 2023.
- **Cloud computing** — IaaS/PaaS/SaaS; MeghRaj (Govt cloud).
- **Digital Twin** — virtual replica for simulation.

### 📡 Digital India Programme (2015)

Nine pillars, most-quoted flagships:

- **UIDAI / Aadhaar** (2009).
- **UPI** (2016) — unified payments; **>15 billion transactions/month** in 2026.
- **DigiLocker, eSanjeevani, CoWIN, DIKSHA, UMANG, BHIM, Rupay, ONDC.**
- **Bharat Net / National Broadband** — fibre to every panchayat.
- **PMGDISHA** — rural digital literacy.

### Exam pointers

- "UPI operator?" → **NPCI (National Payments Corporation of India).**
- "IndiaAI budget?" → **₹10,371 crore, approved March 2024.**
- "National Quantum Mission corpus?" → **₹6,003 crore (2023-31).**
- "BHASHINI covers?" → **22 scheduled Indian languages.**
- "CBDC / Digital Rupee pilot launched?" → **Dec 2022 (retail).**

### In simple words

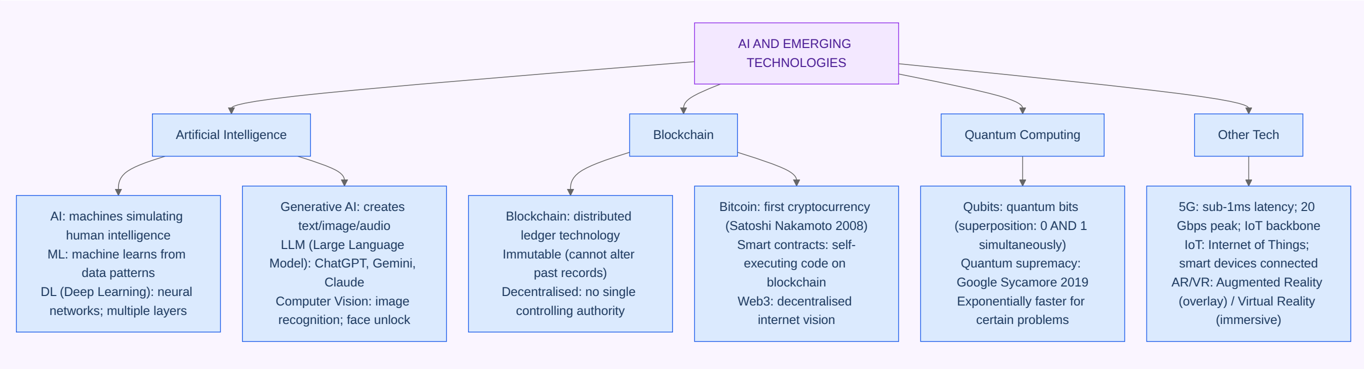
AI today is basically **pattern-matching on steroids**. Feed a machine millions of cat pictures → it learns "cat-ness" and flags new cats. Feed it trillions of words → it learns to predict the next word (that's ChatGPT). No consciousness, no magic — just math + data + GPUs. India's challenge: build our own models in our own languages so we're not renting tech from California.

### Practice

1. List 3 differences between Narrow AI and AGI.
2. Explain UPI in a minute; who operates it, what year it launched, who regulates it.
3. What is the National Quantum Mission's budget and target qubit count?

# PART E — BIOTECHNOLOGY & HEALTH TECH

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW




## Chapter E1 — Genetic Engineering Primer

### 🧬 Core Vocabulary

- **DNA** — double helix, carries hereditary info (A-T, G-C pairs).
- **RNA** — single-strand, transcribed from DNA, translated to protein.
- **Gene** — a DNA segment coding one (or more) protein.
- **Genome** — full DNA set.
- **Genomics** — study of genomes.
- **Proteomics** — study of proteins.

### 🔪 Key Techniques

- **PCR** (Polymerase Chain Reaction) — amplify DNA. Kary Mullis, Nobel 1993. COVID testing workhorse.
- **Recombinant DNA** — cut-paste genes across organisms (enables insulin, hormones).
- **CRISPR-Cas9** — gene-editing "molecular scissors". Charpentier & Doudna, Nobel 2020.
- **Stem cells** — undifferentiated cells (embryonic / adult / iPSC).
- **Cloning** — Dolly the sheep (1996, first mammal).

### 💉 Vaccines — India's Global Role

- India = **pharmacy of the world** (>60% of world's vaccines by volume).
- **Serum Institute** (Pune) — world's largest by volume.
- **Bharat Biotech** — Covaxin, Rotavac, iNCOVACC (nasal COVID vaccine).
- **COVID-era launches (Jan 2021):**
- **Covishield** (Oxford-AstraZeneca, made by Serum).
- **Covaxin** (Bharat Biotech — India's 1st indigenous COVID vaccine, inactivated virus).
- **ZyCoV-D** (Zydus, world's 1st DNA-plasmid COVID vaccine, needle-free).
- **CoWIN** platform — vaccination registry; open-sourced, used by other countries.

### 🌾 Agri-Biotech

- **Bt cotton** — first GM crop approved in India (2002).

- **Bt brinjal** — GEAC cleared (2010) but moratorium imposed.
- **DMH-11 mustard** (2022 GEAC clearance) — GM mustard, under SC scrutiny.
- **Golden Rice** — Vitamin-A-fortified rice (international).
- **GEAC** (Genetic Engineering Appraisal Committee) — regulator under MoEFCC.

### 🧪 Regulatory Bodies

- **DBT** (Department of Biotechnology) — under MoS&T.
- **BIRAC** — Biotechnology Industry Research Assistance Council (funding agency).
- **NII / NCBS / THSTI / IGIB / inStem / RCB** — flagship biotech labs.
- **ICMR** — medical research apex body.

### Exam pointers

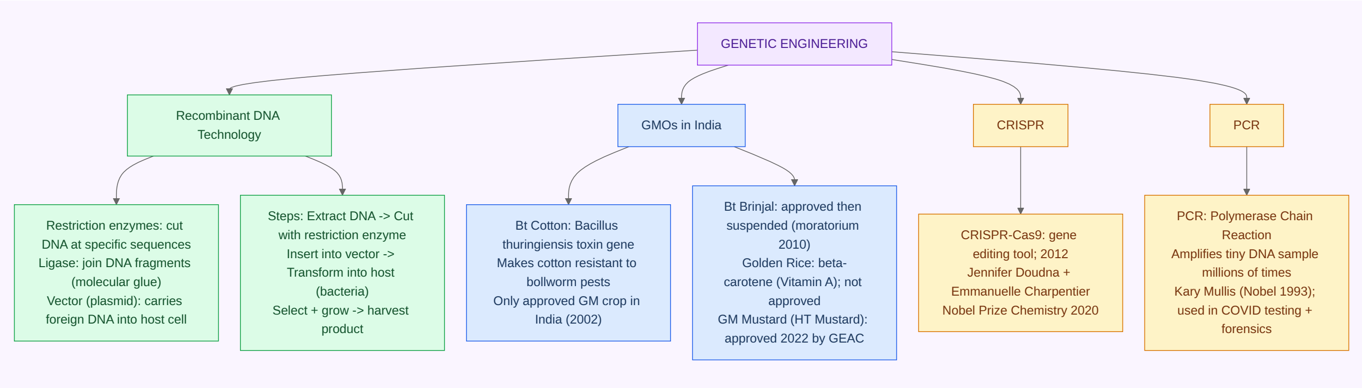
- "Bt cotton approved in India?" → **2002 (first GM crop).**
- "CRISPR Nobel?" → **Charpentier & Doudna, 2020.**
- "Indigenous COVID vaccine?" → **Covaxin (Bharat Biotech).**
- "World's 1st DNA-plasmid COVID vaccine?" → **ZyCoV-D (Zydus, India).**
- "GM mustard = ?" → **DMH-11.**

### Practice

1. Trace insulin production: from gene to product, using recombinant DNA.
2. Why is CRISPR a revolution over older gene-editing methods?
3. List three indigenous vaccines developed by Bharat Biotech.

# PART F — ENERGY & ENVIRONMENT TECH

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW




## Chapter F1 — India's Energy Transition

### 🌞 Renewable Milestones (Apr 2026)

- **Installed power capacity:** ~450 GW.
- **Non-fossil share:** ~47% (target 50% by 2030).
- **Solar:** ~93 GW.
- **Wind:** ~46 GW.
- **Large hydro:** ~47 GW.
- **Nuclear:** ~7.5 GW.

### Paris Pledges (NDCs, updated 2022)

1. 50% non-fossil capacity by 2030.
2. Reduce emissions intensity of GDP by 45% (from 2005 levels) by 2030.
3. Additional **2.5-3 billion tonnes $CO_2$** forest sink by 2030.
4. **Net-Zero by 2070** (Glasgow COP26).

### 🛰️ Key Missions

- **International Solar Alliance** (2015, with France) — 100+ members.
- **ISA Summit 2018**, HQ Gurugram.
- **PM-KUSUM** — solar pumps for farmers.
- **PLI Scheme — solar modules, batteries.**
- **National Green Hydrogen Mission** (Jan 2023, ₹19,744 cr) — target 5 MMT green hydrogen by 2030.
- **National Electric Mobility Mission (NEMMP)** + **FAME-I & II**.

### ⚡ Grid & Transmission

- **One Nation One Grid** achieved 2013 (Southern grid synced).
- **POSOCO → Grid Controller of India Ltd.**
- **Green energy open access rules 2022** — simpler RE procurement.

### Exam pointers

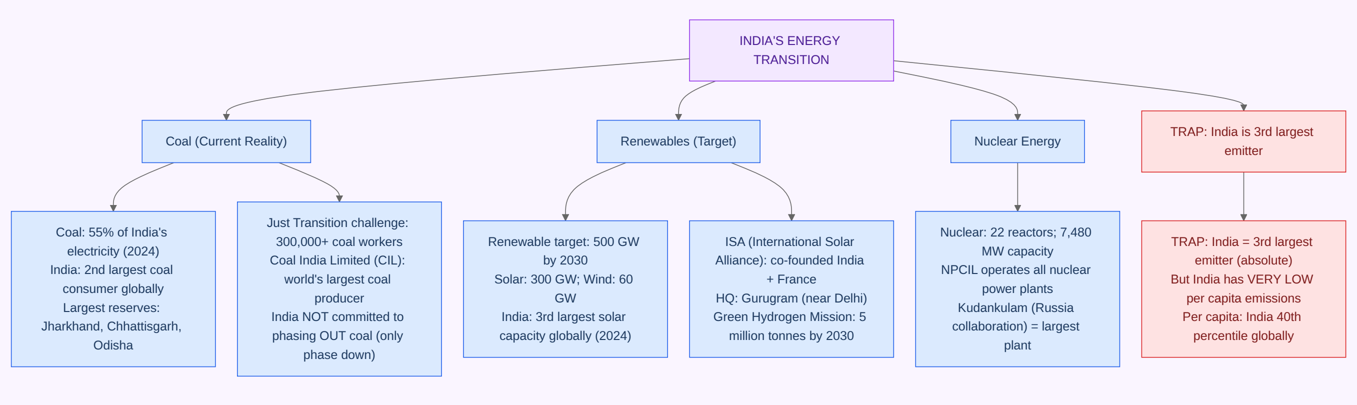
- "India's net-zero target?" → **2070.**
- "ISA HQ?" → **Gurugram (Haryana).**
- "National Green Hydrogen Mission outlay?" → **₹19,744 crore.**
- "Non-fossil share target 2030?" → **50% installed capacity.**

### Practice

1. What are India's four NDC targets?
2. Why is green hydrogen called "green" vs grey/blue?
3. Name three flagship solar schemes.

# PART G — HEALTH & MEDICAL TECH

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW




## Chapter G1 — Medical Missions & Innovations

### 🩺 Key Programmes

- **Ayushman Bharat PM-JAY** (2018) — world's largest public health insurance (₹5 lakh/family/year; 55 crore beneficiaries).
- **Ayushman Bharat Digital Mission (ABDM)** — ABHA health ID; health records stack.
- **Mission Indradhanush** — universal immunisation for children & pregnant women.
- **eSanjeevani** — national telemedicine; 10+ crore consultations.
- **CoWIN** — COVID vaccination backbone.
- **PMBJP** — Jan Aushadhi Pariyojana (generic medicines).
- **PM-ABHIM** — PM Ayushman Bharat Health Infrastructure Mission (₹64,180 cr).

### 🧪 Indigenous Innovations

- **Indian HPV vaccine (CERVAVAC)** — Sep 2022, Serum Institute.
- **iNCOVACC** — intranasal COVID booster (world's first).
- **Polio eradication certification** — India declared polio-free in 2014 by WHO.
- **Tele-MANAS** — National Tele Mental Health Programme (2022).

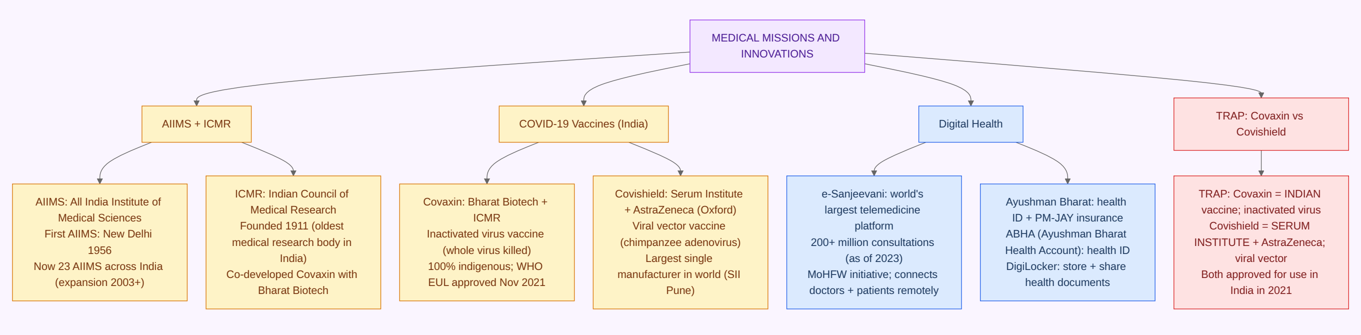
### Exam pointers

- "PM-JAY coverage?" → **₹5 lakh/family/year.**
- "World's 1st intranasal COVID vaccine?" → **iNCOVACC.**
- "India declared polio-free?" → **2014.**
- "Indigenous HPV vaccine?" → **CERVAVAC (2022).**

# PART H — PHYSICS, CHEMISTRY, BIOLOGY ESSENTIALS

*Physics, Chemistry, and Biology each have dedicated full-length books in this series. This part provides a rapid-revision layer for the general-awareness flavour of questions that appear in SSC and Banking exams.*

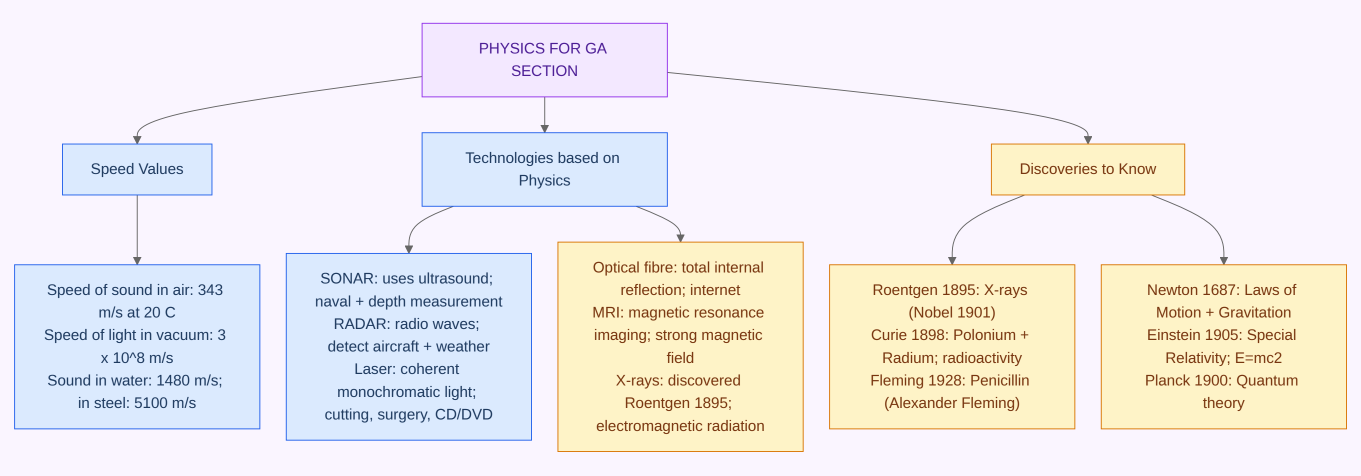
◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW




## Chapter H1 — Physics Essentials (General Awareness flavour)

- **SI units** — 7 base: m, kg, s, A, K, mol, cd.
- **Laws of Motion** (Newton) — 1st (inertia), 2nd (F = ma), 3rd (action-reaction).
- **Laws of Thermodynamics** — 0th (thermal equilibrium), 1st (energy conservation), 2nd (entropy), 3rd (absolute zero unreachable).
- **Electromagnetic spectrum** (low → high frequency): **Radio → Microwave → Infrared → Visible → UV → X-rays → Gamma**.
- **Visible light (VIBGYOR)** — 380-750 nm.
- **Speed of light in vacuum** — $3\times10^8$ m/s.
- **Sound** — longitudinal, ~343 m/s in air, ~1500 m/s in water, faster in solids.
- **LASER** — Light Amplification by Stimulated Emission of Radiation.

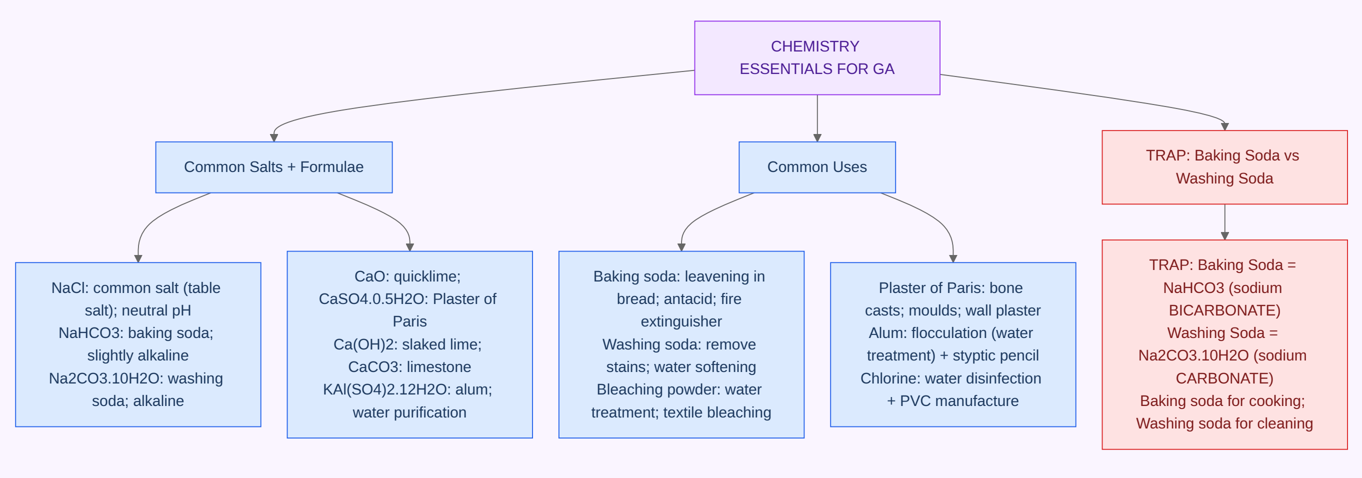
◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



## Chapter H2 — Chemistry Essentials

- **Periodic Table** — 118 elements; **Mendeleev** (1869) → modern by Moseley (atomic number).
- **S-block, p-block, d-block (transition), f-block (lanthanides+actinides).**
- **Atomic structure** — Dalton → Thomson (plum pudding) → Rutherford → Bohr → Quantum.
- **Acids/Bases** — Arrhenius, Brønsted-Lowry, Lewis.
- **pH** — 0-14; $< 7$ acid, $> 7$ base.
- **Key gases & uses:** $O_2$ (respiration), $CO_2$ (photosynthesis/fire-ext), $N_2$ (fertiliser, inert), $H_2$ (fuel of future), He (balloons), Ne (sign lights), Ar (bulb filling).

◎ **CHAPTER SUMMARY — EVERYTHING IN ONE VIEW**



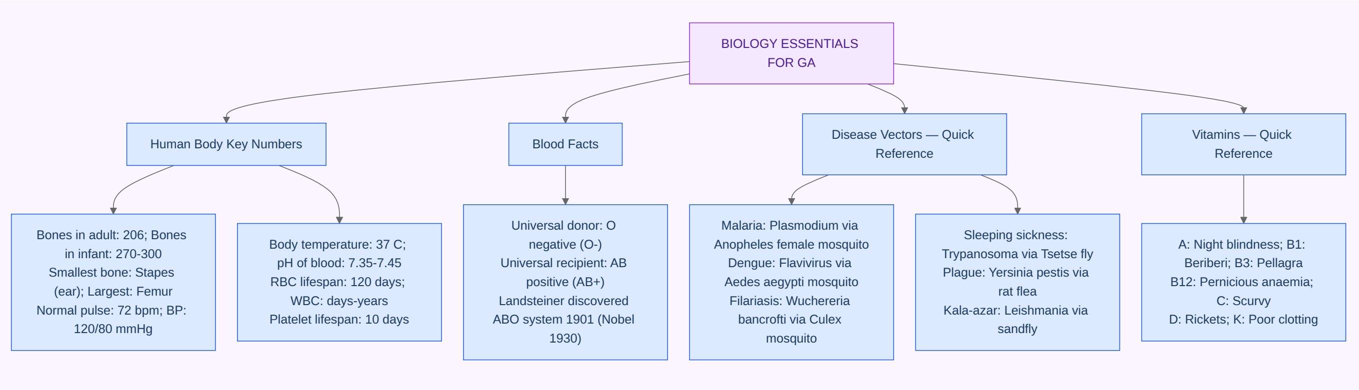

## Chapter H3 — Biology Essentials

- **Cell** = basic unit of life.
- **Prokaryote** (no nucleus — bacteria) vs **Eukaryote** (plants, animals, fungi, protists).
- **Five kingdoms** (Whittaker 1969): Monera, Protista, Fungi, Plantae, Animalia.
- **11 human systems** — integumentary, skeletal, muscular, nervous, endocrine, cardiovascular, lymphatic, respiratory, digestive, urinary, reproductive.
- **Blood groups** — A, B, AB, O × Rh+/−. AB+ universal recipient, O− universal donor.
- **Vitamins deficiency:** A (night blindness), $B_1$ (beriberi), $B_3$ (pellagra), $B_{12}$ (anaemia), C (scurvy), D (rickets), E (sterility), K (bleeding).

📎 ***For depth, jump to the dedicated subject books.***

# APPENDICES

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW




## Appendix 1 — One-page cheatsheet

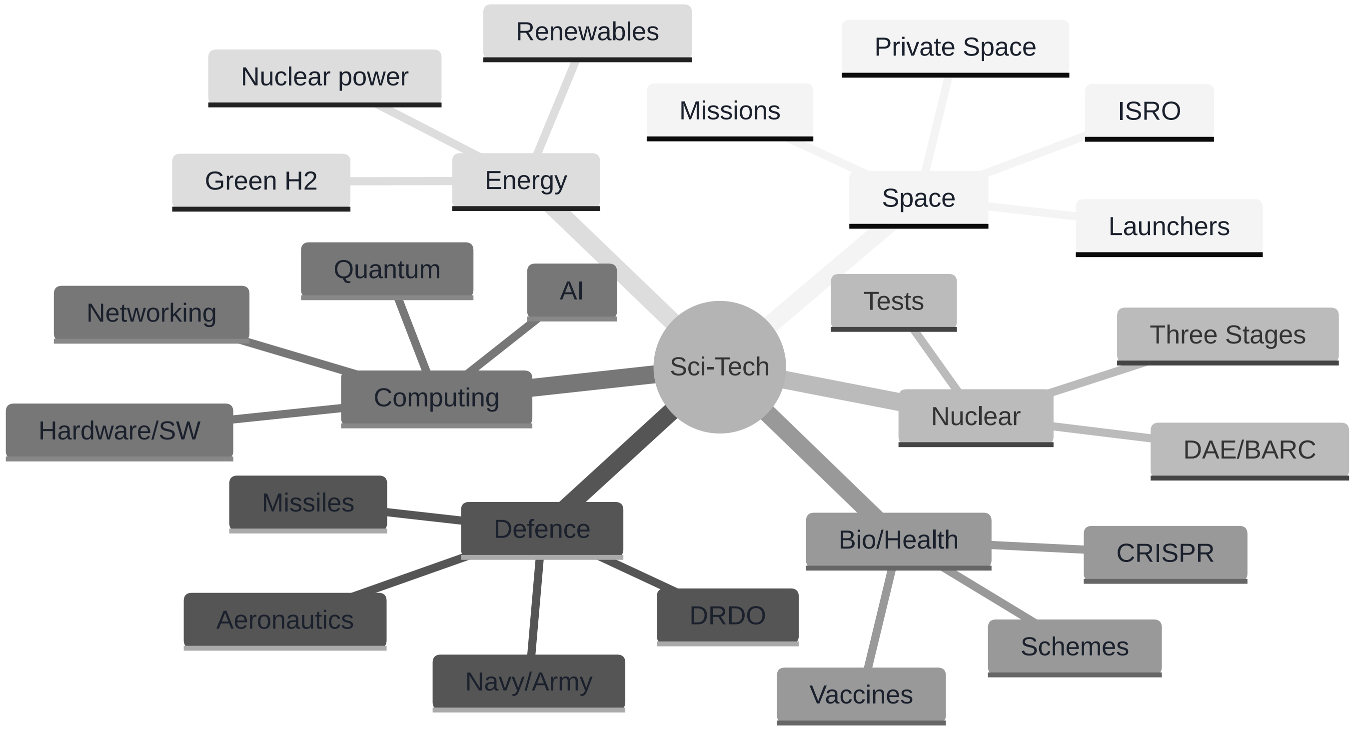
| Category | Key Fact |
|---|---|
| ISRO HQ | Bengaluru |
| DRDO HQ | New Delhi |
| DAE HQ | Mumbai |
| AEC/BARC | Mumbai |
| PSLV record | 104 sats (2017) |
| Chandrayaan-3 landing | 23 Aug 2023 |
| National Space Day | 23 August |
| National Technology Day | 11 May |
| Mission Shakti (ASAT) | 27 Mar 2019 |
| INS Vikrant | Commissioned Sep 2022 |
| Agni-V MIRV | Mission Divyastra, 2024 |
| First indigenous COVID vaccine | Covaxin |
| Nuclear 3-stage | PHWR → FBR → Thorium |
| Net-zero year | 2070 |
| IndiaAI Mission corpus | ₹10,371 cr |
| NQM corpus | ₹6,003 cr (2023-31) |

## Appendix 2 — Master Mnemonics List

- **PATAN** → Prithvi-Agni-Trishul-Akash-Nag (IGMDP).

- **Launchpad Walk** → SSLV-PSLV-GSLV-LVM3-Gaganyaan.
- **PDNTSPA** (Please Do Not Throw Sausage Pizza Away) → OSI 7 layers.
- **VIBGYOR** → Violet-Indigo-Blue-Green-Yellow-Orange-Red.

## Appendix 3 — Mind Map (Sci-Tech)



# PART X — RECENT ISRO MISSIONS (2020-2026)

| Year | Mission | Launcher | Notes |
| --- | --- | --- | --- |
| 2019 (Jul) | **Chandrayaan-2** | GSLV Mk-III M1 | Orbiter, Vikram lander (crashed), Pragyan rover |
| 2020 (Nov) | EOS-01 (RISAT-2BR2) | PSLV-C49 | radar imaging |
| 2021 (Feb) | Amazonia-1 (Brazil) + 18 co-passengers | PSLV-C51 | first dedicated commercial PSLV under NSIL |
| 2022 (Aug) | **SSLV-D1** (maiden flight) | SSLV (1st) | EOS-02 partial failure |
| 2022 (Oct) | OneWeb 36 satellites | LVM3 (rebranded GSLV Mk-III) | first commercial LVM3 |
| 2023 (Mar) | OneWeb 36 satellites (2nd batch) | LVM3 | |
| 2023 (Jul) | **Chandrayaan-3** | LVM3 M4 | **soft landing 23 Aug 2023** at Shiv Shakti Point near south pole — India 4th country (after USA, USSR, China); 1st to land near lunar south pole |
| 2023 (Sep) | **Aditya-L1** | PSLV-C57 | India's 1st solar mission; placed at Lagrange-1 in Jan 2024 |
| 2023 (Oct) | Gaganyaan TV-D1 | Test Vehicle | crew-escape test (in-flight abort) |
| 2024 (Jan) | XPoSat (X-ray Polarimeter Satellite) | PSLV-C58 | studies cosmic X-ray polarisation |
| 2024 (Feb) | INSAT-3DS | GSLV F14 | weather + meteorology |
| 2024 (Aug) | EOS-08 | SSLV-D3 | 3rd & final SSLV development flight; SSLV now operational |
| 2024 (30 Dec) | **SpaDeX** (Space Docking Experiment) | PSLV-C60 | India's 1st space-docking mission, **docking achieved 16 Jan 2025** — 4th country (after USA, Russia, China) |
| 2025 (Jul) | **NISAR** (NASA-ISRO SAR) — launched 30 Jul 2025 | GSLV-F16 | joint Earth-observation Synthetic Aperture Radar mission |
| **2026 (H2)** | **Gaganyaan-1 (uncrewed) — Vyommitra robot onboard** | LVM3 | First uncrewed orbital test (delayed from 2025); will simulate full human spaceflight profile |
| **2027 (H1)** | **Gaganyaan crewed mission (G2 + G3 uncrewed first)** | LVM3 | India's first human spaceflight |

| Year | Mission | Launcher | Notes |
|---|---|---|---|
| 2026-28 (planned) | **Chandrayaan-4** | LVM3 + PSLV (split) | sample-return mission |
| 2028 (planned) | **Mangalyaan-2 (MOM-2)** | LVM3 | second Mars mission |
| 2028 (planned) | **Shukrayaan-1** (Venus mission) | LVM3 | Venus orbiter |
| 2035 (vision) | **Bharatiya Antariksh Station** | — | India's space station |
| 2040 (vision) | Indian on Moon | — | crewed lunar mission |

## ISRO launch vehicle evolution

| Vehicle | Years active | Payload to LEO | Notes |
|---|---|---|---|
| SLV-3 | 1979-83 | 40 kg | India's first; Rohini satellites |
| ASLV | 1987-94 | 150 kg | augmented SLV |
| **PSLV** | 1993- | 1,750 kg (sun-sync) | "workhorse"; XL variant carries up to 1.9 t |
| **GSLV Mk-II** | 2001- | 2,500 kg (LEO), 2,200 kg (GTO) | indigenous cryogenic stage |
| **LVM3** (GSLV Mk-III) | 2014- | 8,000 kg LEO, 4,000 kg GTO | Chandrayaan-2/3, Gaganyaan |
| **SSLV** | 2022- | 500 kg | small-satellite launch; on-demand |
| Reusable Launch Vehicle (RLV-TD) | 2016 (test), 2024 (REX-3) | demonstration | autonomous landing tests |
| **NGLV (Soorya)** | 2031 (planned) | 10-15 t LEO | next-gen heavy-lift, partially reusable |

## Indian astronauts (Vyomanaut programme)

- **Rakesh Sharma** — first Indian in space (1984) on Soviet Soyuz T-11.
- **Wing Commander Subhash Chandra Pradeep Krishnan Nair (Prashanth Nair), Group Captain Ajit Krishnan, Group Captain Angad Pratap, Wing Commander Shubhanshu Shukla** — Gaganyaan-1 designate astronauts (announced Feb 2024).

# PART Y — DEFENCE TECHNOLOGY (recent)

## Missiles — operational (DRDO)

| Missile | Type | Range |
|---|---|---|
| Prithvi-I/II/III | SRBM (surface-to-surface) | 150-350 km |
| Agni-I | SRBM | 700 km |
| Agni-II | MRBM | 2,000 km |
| Agni-III | IRBM | 3,500 km |
| Agni-IV | IRBM | 4,000 km |
| **Agni-V** | ICBM (with MIRV — Mission Divyastra Mar 2024) | 5,000+ km |
| **Agni-Prime (Agni-P)** | new-gen MRBM | 1,000-2,000 km |
| Trishul | short-range SAM | 9 km (decommissioned) |
| Akash | medium-range SAM | 25-30 km |
| **Akash-NG** | new-gen SAM | 70-80 km |
| Nag / Helina | anti-tank | 4-7 km |
| Nag (NAMICA) | vehicle-mounted | 7 km |
| **Astra Mk-I/Mk-II/Mk-III** | beyond-visual-range air-to-air | 110 / 160 / 350 km |
| **Rudram-1** | anti-radiation missile | 100-250 km |
| **BrahMos** | supersonic cruise | 290 → 450 → 800 km variants |
| BrahMos-NG | smaller, lighter variant | 290-500 km |
| **BrahMos-II** (planned) | hypersonic | 1,500 km |
| K-15 (Sagarika) | SLBM | 700 km |
| **K-4** | SLBM | 3,500 km |
| K-5 (planned) | SLBM | 5,000 km |
| **Pinaka Mk-I/II** | multi-barrel rocket launcher | 38-90 km |
| Pinaka-ER | extended range | 75 km |
| **Pralay** | quasi-ballistic | 150-500 km |
| Shaurya | hypersonic | 750 km |

## Recent indigenous defence platforms

- **INS Vikrant** — India's first indigenous aircraft carrier; commissioned Sep 2022 at Cochin Shipyard.
- **INS Arihant** — India's first indigenous nuclear-powered ballistic-missile submarine (SSBN) — completed nuclear triad. INS Arighat, INS Aridaman to follow.
- **INS Vagir, Vagsheer** — Project-75 Scorpene-class diesel-electric submarines.
- **HAL Tejas Mk-1A** — light combat aircraft; Mk-2 (medium combat) under development.
- **TAPAS BH-201** — DRDO's medium-altitude long-endurance UAV.
- **Rustom-II / TAPAS** — DRDO MALE UAV.
- **AMCA** (Advanced Medium Combat Aircraft) — 5th-gen stealth fighter under development.
- **Arjun Mk-1A** — main battle tank (DRDO + HVF Avadi).
- **Dhanush, Sharang** — artillery guns (155 mm).
- **K-9 Vajra-T** — self-propelled artillery (L&T + Hanwha).
- **NETRA** — DRDO airborne early-warning + control (AEW&C); follow-on Netra Mk-1A, Mk-2.

## Recent / repeat military exercises (high-yield)

| Exercise | Partner | Type |
|---|---|---|
| Yudh Abhyas | USA | Army |
| Vajra Prahar | USA | Special Forces |
| Cope India | USA | Air |
| Malabar | USA + Japan + Australia | Naval (QUAD) |
| Tiger Triumph | USA | Tri-services amphibious |
| Mitra Shakti | Sri Lanka | Army |
| SLINEX | Sri Lanka | Naval |
| Indra | Russia | Tri-services |
| Hand-in-Hand | China | Army (rare; suspended post-Galwan) |
| JIMEX | Japan | Naval |
| Dharma Guardian | Japan | Army |
| Ekuverin | Maldives | Army |
| Surya Kiran | Nepal | Army |
| Nomadic Elephant | Mongolia | Army |
| Garuda | France | Air |
| Varuna | France | Naval |

| Exercise | Partner | Type |
|---|---|---|
| Shakti | France | Army |
| Konkan | UK | Naval |
| Ajeya Warrior | UK | Army |
| Indradhanush | UK | Air |
| AUSINDEX | Australia | Naval |
| AUSTRA HIND | Australia | Army |
| Sampriti | Bangladesh | Army |
| Bongosagar | Bangladesh | Naval |
| Maitree | Thailand | Army |
| Dosti | Maldives + Sri Lanka | Coast Guard |
| Vinbax | Vietnam | UN engineering |
| Khanjar | Kyrgyzstan | Special Forces |
| Tarkash | USA NSG | counter-terror |
| Cobra Warrior | UK | multilateral air |
| Rim of the Pacific (RIMPAC) | 25+ nations | naval (USA-led) |
| Red Flag | USA | air (multilateral) |
| **Tarang Shakti** | 30 nations | India's largest multilateral air exercise (2024 first edition) |

## Defence procurement schemes

- **Make in India** (Defence) — emphasising indigenous manufacture.
- **iDEX** (Innovations for Defence Excellence) — DRDO + DPSU + start-ups.
- **Strategic Partnership model** — Indian companies tying up with foreign OEMs.
- **Negative Imports List** — items that can ONLY be procured indigenously (5 lists released 2020-2024).

# PART Z — COMPUTER FUNDAMENTALS DEEP-DRILL

## Z.1 Generations of computers

| Generation | Period | Key device |
|---|---|---|
| 1st | 1946-59 | vacuum tubes (ENIAC, UNIVAC) |
| 2nd | 1959-65 | transistors (IBM 1401, 7094) |
| 3rd | 1965-71 | integrated circuits (IBM 360) |
| 4th | 1971-present | VLSI / microprocessors (Intel, Apple, IBM PC) |
| 5th | future | AI, ULSI, parallel processing, quantum |

## Z.2 Computer architecture (Von Neumann)

- **CPU** = Control Unit (CU) + Arithmetic Logic Unit (ALU) + Registers.
- **Memory hierarchy** (fast → slow):
- Registers (in CPU, ns)
- Cache (L1, L2, L3 — SRAM)
- Main Memory (RAM — DRAM)
- Secondary storage (HDD, SSD)
- Tertiary (optical, tape)
- **Buses** — Address (where), Data (what), Control (how).

## Z.3 Memory units

| Unit | Bytes |
|---|---|
| 1 nibble | 4 bits |
| 1 byte | 8 bits |
| 1 kilobyte (KB) | 1,024 B |
| 1 megabyte (MB) | 1,024 KB |
| 1 gigabyte (GB) | 1,024 MB |
| 1 terabyte (TB) | 1,024 GB |
| 1 petabyte (PB) | 1,024 TB |
| 1 exabyte (EB) | 1,024 PB |
| 1 zettabyte (ZB) | 1,024 EB |
| 1 yottabyte (YB) | 1,024 ZB |

## Z.4 Number systems

| System | Base | Digits |
|---|---|---|
| Binary | 2 | 0, 1 |
| Octal | 8 | 0-7 |
| Decimal | 10 | 0-9 |
| Hexadecimal | 16 | 0-9, A-F |

- ASCII — 7 bits (128 chars); extended ASCII 8 bits (256).
- UTF-8 — variable-width Unicode, ASCII-compatible.
- BCD — Binary Coded Decimal (4 bits per digit).

## Z.5 Software taxonomy

- **System software** — OS, device drivers, utility programs, language translators (assembler / compiler / interpreter).
- **Application software** — word processors, spreadsheets, browsers, databases.
- **Open-source** — Linux, Apache, MySQL, Python, Firefox.
- **Proprietary** — Windows, MS Office, Photoshop.

## Z.6 Operating systems

- **Linux** (Linus Torvalds, 1991), Ubuntu, Debian, Red Hat, Fedora, Arch.
- **Windows** — Microsoft (1985-).
- **macOS** — Apple (Unix-based).
- **Android** — Google (Linux-based; mobile).
- **iOS** — Apple (mobile).
- **Chrome OS** — Google (browser-based).
- **BharatOS (BHRAT-OS)** — IIT Madras + JandKops; Indian mobile OS.

## Z.7 Networking — OSI 7 layers

| Layer | Function | Example protocol |
|---|---|---|
| 7. Application | end-user services | HTTP, FTP, SMTP, DNS, SSH |
| 6. Presentation | encoding, encryption | SSL/TLS, JPEG, ASCII |
| 5. Session | session management | NetBIOS, RPC |
| 4. Transport | reliability, flow control | TCP, UDP |
| 3. Network | routing | IP, ICMP, OSPF, BGP |
| 2. Data Link | frames, MAC | Ethernet, PPP, ARP |
| 1. Physical | bits over wire | Cables, hubs, RJ-45 |

***Mnemonic** (top to bottom): "**A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing".*

## Z.8 TCP/IP suite (4 layers)

| TCP/IP layer | OSI equivalents |
|---|---|
| Application | 7 + 6 + 5 |
| Transport | 4 |
| Internet | 3 |
| Network access | 2 + 1 |

## Z.9 Network terms

- **LAN** — Local Area Network (within building).
- **WAN** — Wide Area Network (continents — internet).
- **MAN** — Metropolitan Area Network (city).
- **PAN** — Personal Area Network (Bluetooth).
- **VPN** — Virtual Private Network (encrypted tunnel).
- **VLAN** — Virtual LAN (logical segmentation).
- **IP address** — IPv4: 32-bit (4.3 billion addresses); IPv6: 128-bit ($3.4 \times 10^{38}$).
- **MAC address** — 48-bit hardware address (e.g., 00:1A:2B:3C:4D:5E).
- **DNS** — Domain Name System; resolves names to IPs.
- **DHCP** — Dynamic Host Configuration Protocol; assigns IPs.
- **NAT** — Network Address Translation.
- **HTTP** — port 80; HTTPS — port 443; SSH — 22; FTP — 21; SMTP — 25; DNS — 53.

## Z.10 Internet milestones

| Year | Event |
|---|---|
| 1969 | ARPANET launched (DARPA, USA) |
| 1971 | First email (Ray Tomlinson) — `@` sign chosen |
| 1983 | TCP/IP becomes standard |
| 1989 | World Wide Web proposed (Tim Berners-Lee at CERN) |
| 1990 | First web browser ("WorldWideWeb" by Berners-Lee) |
| 1993 | Mosaic browser |
| 1995 | Internet commercial in India (VSNL on 15 Aug) |
| 1998 | Google founded (Larry Page + Sergey Brin) |
| 2004 | Facebook founded (Mark Zuckerberg) |

| Year | Event |
|---|---|
| 2005 | YouTube founded |
| 2006 | Twitter founded |
| 2007 | iPhone launched |
| 2008 | Android released |
| 2010 | Instagram founded |
| 2011 | IPv6 launch |
| 2017 | Reliance Jio rolls out 4G in India |
| 2022 | 5G commercial launch in India (Oct, by Airtel + Jio) |
| 2022 | ChatGPT launched (Nov 30, OpenAI) |
| 2023 | Threads, Bard, Gemini, Llama, Claude — generative-AI mainstream |
| 2024 | India crosses **1 billion internet users**; UPI cross-border live in 7 countries |

## Z.11 AI / ML / blockchain (basics)

- **AI** = systems performing tasks that normally need human intelligence.
- **Narrow AI** — excels at one task (chess, image classification).
- **General AI (AGI)** — human-level breadth (still hypothetical).
- **Machine Learning** = AI subfield; learns patterns from data.
- **Supervised** — labelled data (regression, classification).
- **Unsupervised** — unlabelled data (clustering, PCA).
- **Reinforcement** — reward-based learning (game-playing, robotics).
- **Deep Learning** = ML using multi-layer neural networks (CNN for images, RNN/LSTM/Transformer for sequences).
- **Generative AI** — produces new content (text, image, audio, video). Models: GPT (OpenAI), Gemini (Google), Claude (Anthropic), Llama (Meta).
- **Blockchain** = distributed immutable ledger; basis of cryptocurrency.
- **Bitcoin** (Satoshi Nakamoto, 2009) — first cryptocurrency.
- **Ethereum** — programmable smart contracts.
- **Web3** — decentralised internet built on blockchain.
- **NFT** — non-fungible token; unique digital asset.
- **Quantum computing** — uses qubits (superposition, entanglement); IBM, Google, IonQ leaders.

## Z.12 Indian IT-government schemes

- **Digital India** (2015) — pillars: digital infrastructure, governance + services on demand, digital empowerment.
- **BharatNet** — fibre to gram panchayats (NOFN).

- **Aadhaar** — UIDAI; world's largest biometric ID (>1.3 billion).
- **DigiLocker** — cloud document storage; Aadhaar-linked.
- **eSign** — Aadhaar-based electronic signature.
- **e-Sanjeevani** — telemedicine.
- **DIKSHA** — teacher + student learning platform.
- **CoWIN** — vaccine registration platform; open-sourced globally.
- **Aarogya Setu** — COVID contact-tracing app.
- **mPassport Seva, mAadhaar, mParivahan** — m-governance.
- **GeM (Government e-Marketplace)** — public procurement portal.
- **IndiaAI Mission** — ₹10,371 cr; computing infra + datasets + AI startups + IndiaAI dataset platform.
- **National Quantum Mission (NQM)** — ₹6,003 cr; 4 thematic hubs (computing, communication, sensing, materials).

## PART W — BIOTECH RECENT (extension)

- **CRISPR-Cas9** — Jennifer Doudna + Emmanuelle Charpentier (Nobel 2020).
- **mRNA vaccines** — Pfizer/BioNTech, Moderna (2020); Karikó + Weissman (Nobel Medicine 2023) for nucleoside-modified mRNA.
- **Indian COVID vaccines** — Covishield (Serum Institute), Covaxin (Bharat Biotech, indigenous, killed-virus), Corbevax (Biological E), Sputnik V (Russia, manufactured in India), ZyCoV-D (Zydus, world's first DNA vaccine), iNCOVACC (Bharat Biotech, intranasal).
- **Genome India Project** — 10,000-Indian genome sequencing programme (DBT).
- **Gene therapy approvals** — for sickle-cell disease (Casgevy by Vertex/CRISPR Tx, Dec 2023, FDA — first CRISPR-based therapy approved).
- **GM crops in India** — only Bt cotton (commercial since 2002); Bt brinjal under moratorium; GM mustard (DMH-11) approved for environmental release Oct 2022 — but stayed by Supreme Court.

*Parts X–W are particularly current (2024–26) and should be reviewed annually as ISRO/DRDO add new operational systems.*

# PART V — NOBEL PRIZES 2020-2024 (full 5 years)

## Physics

| Year | Laureates | For |
|---|---|---|
| 2020 | Roger Penrose, Reinhard Genzel, Andrea Ghez | Black holes; supermassive black hole at galactic centre |
| 2021 | Syukuro Manabe, Klaus Hasselmann, Giorgio Parisi | Climate models + complex physical systems |
| 2022 | Alain Aspect, John Clauser, Anton Zeilinger | Quantum entanglement experiments + Bell inequalities |
| 2023 | Pierre Agostini, Ferenc Krausz, Anne L'Huillier | Attosecond pulses (electron motion) |
| 2024 | John Hopfield + Geoffrey Hinton | Foundational discoveries in artificial neural networks ("Godfathers of AI") |
| **2025** | **John Clarke + Michel H. Devoret + John M. Martinis** | **Macroscopic quantum-mechanical tunnelling + energy quantisation in superconducting circuits** |

## Chemistry

| Year | Laureates | For |
|---|---|---|
| 2020 | Emmanuelle Charpentier + Jennifer Doudna | **CRISPR-Cas9 gene editing** |
| 2021 | Benjamin List + David MacMillan | Asymmetric organocatalysis |
| 2022 | Carolyn Bertozzi, Morten Meldal, K. Barry Sharpless | Click chemistry + bioorthogonal chemistry (Sharpless 2nd Nobel after 2001) |
| 2023 | Moungi Bawendi, Louis Brus, Alexei Ekimov | Quantum dots |
| 2024 | David Baker (computational protein design) + Demis Hassabis + John Jumper (AlphaFold — predicting protein structure) | Protein structure prediction |
| **2025** | **Susumu Kitagawa (Japan) + Richard Robson (Australia) + Omar M. Yaghi (USA)** | **Metal-organic frameworks (MOFs) — porous materials with applications in gas storage, drug delivery, catalysis** |

## Medicine / Physiology

| Year | Laureates | For |
|---|---|---|
| 2020 | Harvey Alter, Michael Houghton, Charles Rice | Hepatitis C virus discovery |
| 2021 | David Julius + Ardem Patapoutian | Receptors for temperature + touch |

| Year | Laureates | For |
| --- | --- | --- |
| 2022 | Svante Pääbo | Genomes of extinct hominins (Neanderthal); founder of paleogenomics |
| 2023 | Katalin Karikó + Drew Weissman | mRNA vaccines (foundational nucleoside-modified mRNA work) |
| 2024 | Victor Ambros + Gary Ruvkun | microRNA discovery + post-transcriptional gene regulation |
| **2025** | **Mary E. Brunkow + Fred Ramsdell (USA) + Shimon Sakaguchi (Japan)** | **Regulatory T cells — peripheral immune tolerance; how immune system avoids attacking own body** |

## Literature

| Year | Laureate | Country |
| --- | --- | --- |
| 2020 | Louise Glück | USA |
| 2021 | Abdulrazak Gurnah | Tanzania-UK |
| 2022 | Annie Ernaux | France |
| 2023 | Jon Fosse | Norway |
| 2024 | Han Kang | South Korea (first Asian woman) |
| **2025** | **László Krasznahorkai** | **Hungary** — for "compelling and visionary oeuvre that, in midst of apocalyptic terror, reaffirms power of art" |

## Peace

| Year | Laureate | For |
| --- | --- | --- |
| 2020 | World Food Programme (WFP) | Hunger relief |
| 2021 | Maria Ressa + Dmitry Muratov | Press freedom |
| 2022 | Ales Bialiatski (Belarus) + Memorial (Russia) + CCL (Ukraine) | Human rights |
| 2023 | Narges Mohammadi (Iran, in jail) | Women's rights in Iran |
| 2024 | Nihon Hidankyo (Japan) | Hibakusha (atomic bomb survivors); push for nuclear-weapon abolition |
| **2025** | **María Corina Machado (Venezuela)** | **For tireless work promoting democratic rights for the people of Venezuela** |

## Economics

| Year | Laureates | For |
|---|---|---|
| 2020 | Paul Milgrom + Robert Wilson | Auction theory |
| 2021 | David Card, Joshua Angrist, Guido Imbens | Empirical labour economics; natural experiments |
| 2022 | Ben Bernanke, Douglas Diamond, Philip Dybvig | Banks + financial crises |
| 2023 | Claudia Goldin | Women in labour market (only 3rd woman; first solo woman in Economics) |
| 2024 | Daron Acemoglu, Simon Johnson, James Robinson | Why nations succeed/fail (institutions) |
| **2025** | **Joel Mokyr (Israeli-American) + Philippe Aghion (France) + Peter Howitt (Canada)** | **Innovation-driven economic growth** |

# PART W — RECENT (2024-26) SCIENCE & TECH BREAKTHROUGHS

## Space (2024-26)

- **Chandrayaan-3** (Aug 2023) — India 4th nation to soft-land on Moon; 1st near south pole.
- **Aditya-L1** placed at L1 (Jan 2024) — India's first solar mission.
- **XPoSat** (Jan 2024) — India's X-ray polarimeter satellite.
- **SpaDeX** (Dec 2024 launch; docking 16 Jan 2025) — India 4th country to demonstrate space-docking.
- **Gaganyaan TV-D1** (Oct 2023) — crew-escape test.
- **Gaganyaan astronauts announced** (Feb 2024) — Prashanth Nair, Ajit Krishnan, Angad Pratap, Shubhanshu Shukla.
- **NISAR** (launched **30 Jul 2025** by GSLV-F16, per ISRO official) — joint NASA-ISRO Earth-observation Synthetic Aperture Radar satellite.
- **NASA Artemis-2** (planned 2025-26) — first crewed mission around Moon since 1972.
- **SpaceX Starship** test flights (2023-25) — multiple integrated flight tests; world's largest rocket.
- **JWST** (James Webb Space Telescope, launched 2021) — operational; deepest infrared images ever.
- **Euclid telescope** (ESA, launched 2023) — dark matter + dark energy mapping.

## AI / ML / computing

- **ChatGPT** (Nov 2022) — first widely-used LLM; OpenAI; led to ChatGPT-4 (Mar 2023), GPT-4o (May 2024), GPT-5 (mid-2025).
- **DALL-E, Midjourney, Stable Diffusion** — text-to-image generators.
- **Sora, Veo, Runway** — text-to-video.
- **Google Gemini** (Dec 2023) — multimodal; **Gemini 2.0** (Dec 2024).
- **Anthropic Claude 3.5 Sonnet** (Jun 2024) — frontier reasoning model.
- **Llama 3.1** (Meta, Jul 2024) — open-source 405B parameter model.
- **DeepMind AlphaFold-3** (May 2024) — protein-structure prediction with arbitrary biomolecules.
- **NVIDIA H100, H200, B100** — GPUs powering AI training.
- **ChatGPT Search** (Oct 2024) — search integration.

## Biotech / health

- **CRISPR-Cas9** therapy (Casgevy, Vertex/CRISPR Tx, Dec 2023, FDA) — first CRISPR-based therapy for sickle cell + beta-thalassemia.
- **AlphaFold 3** (Google DeepMind, May 2024) — biomolecular structure prediction.
- **GLP-1 receptor agonists** (Ozempic/Wegovy, Mounjaro) — for diabetes + obesity.
- **mRNA flu vaccine** in Phase III (Moderna 2024).
- **Universal cancer vaccine** prototypes (mRNA-based, BioNTech 2025 trials).
- **Anti-aging research** — senolytic drugs in trials.
- **Pig kidney transplant** (Massachusetts General Hospital, Mar 2024) — first to a living human.

- **First womb transplant + birth** in India (AIIMS Delhi 2024).

## Climate + energy

- **Net-zero pledges** — India 2070; China 2060; EU 2050; USA 2050.
- **EV adoption** — India FAME-II + III subsidy schemes; 2030 target 30% new sales electric.
- **Green hydrogen mission** (India, Jan 2023) — ₹19,744 cr; 5 MMT/yr by 2030.
- **Bhadla Solar Park** (Rajasthan) — world's largest solar park (~2.25 GW).
- **Pavagada Solar Park** (Karnataka) — 2.05 GW.
- **First fully indigenous nuclear reactor** — KAPP-3 (Kakrapar, GJ) operational 2023.
- **Fast Breeder Reactor** (Kalpakkam, TN) — operational 2024 (3-stage nuclear program).
- **Fusion progress** — National Ignition Facility (USA) achieved net energy gain (Dec 2022; replicated 2024).
- **ITER** (France, intl) — under construction; first plasma 2025-2026 target.

## Materials + electronics

- **Semiconductor manufacturing in India** — Tata-PSMC fabs (GJ); Micron OSAT (GJ); CG Power (GJ).
- **2 nm chip** mass production (TSMC, 2025).
- **Quantum computing** — IBM 1000-qubit Condor (Dec 2023); Google Willow chip (Dec 2024) — quantum error-correction milestone.
- **Room-temperature superconductor** (LK-99, claimed July 2023) — discredited but research continues.

## Defence

- **Mission Divyastra** (Mar 2024) — Agni-V with MIRV (Multiple Independently-targetable Reentry Vehicle) tested.
- **ET-LDHCM (Extended Trajectory-Long Duration Hypersonic Cruise Missile)** — DRDO tested 14 Jul 2025; **Mach 8** (~9,800 km/h) speed.
- **Scramjet long-duration ground test** (9 Jan 2026) — over 12 minutes; powers hypersonic missile programme.
- **Vertically-Launched SR-SAM** flight-test (26 Mar 2025, Indian Navy).
- **MRSAM Army version** — 4 successful flight-tests (3-4 Apr 2025) from APJ Abdul Kalam Island, Odisha.
- **MPATGM (Man-Portable Anti-Tank Guided Missile)** — top-attack capability test (11 Jan 2026).
- **Agni-V test** (Aug 2025) — range >5,000 km, from Odisha.
- **INS Vikrant** (Sep 2022) — first indigenous aircraft carrier.
- **INS Arighat** (Aug 2024) — second nuclear submarine (Arihant class).
- **HAL Tejas Mk-1A** delivery (2024).
- **AMCA** (Advanced Medium Combat Aircraft) — 5th-gen stealth fighter; design phase complete 2024.
- **Akash NG** SAM operational (2024).
- **K-4 SLBM** test (Jan 2025) — 3,500 km range.
- **Tarang Shakti** (Aug 2024) — India's largest multilateral air exercise (30 nations).

## Indian computer-science / digital

- **5G nationwide rollout** completed 2024 (Reliance Jio + Airtel + Vi). **India = 2nd largest 5G user base globally (after China) — 394 million 5G users by end-2025**; mid-band 5G covers >90% of population.
- **6G research** — TSDSI 6G Vision document (2023); Bharat 6G Mission (2023, ₹1 trillion outlay); **Bharat 6G Alliance now 84+ members**; 6G Apex Council met 10 Dec 2025.
- **Indigenous 4G stack** rolled out by PM Modi on 27 Sep 2025 — **India 5th country in world** with own 4G stack (upgradable to 5G).
- **Internet users milestone** — India crossed **1 billion broadband subscribers in November 2025** (TRAI; 100.29 crore; up from 25.15 crore in March 2014).
- **CBDC retail** (e₹) live in 9 cities; programmable + offline pilots.
- **DigiYatra** at 28+ airports (face-recognition boarding).
- **Aadhaar** crosses 1.4 billion enrolments (2024).
- **UPI** crosses 18-20 billion txns/month (2026); UPI live in 7+ countries.

## Health (India)

- **Ayushman Bharat (PM-JAY)** — covers 50+ crore citizens with ₹5L cover.
- **Jan Aushadhi** — 14,000+ stores (generic medicines).
- **eSanjeevani** telemedicine — 200+ million consultations.
- **iNCOVACC** — Bharat Biotech intranasal COVID vaccine.
- **ZyCoV-D** (Zydus, Aug 2021) — world's first DNA vaccine.

## PART U — SCI-TECH MINI-MOCK (25 Questions · 25 Minutes)

Set a timer. No looking back. Check the answer key at the end.

**1.** Chandrayaan-3's Vikram lander touched down on the Moon on:

(a) 14 July 2023 (b) 23 August 2023 (c) 5 September 2023 (d) 1 January 2024

**2.** India's Aditya-L1 solar mission was positioned at which Lagrange point?

(a) L1 (b) L2 (c) L4 (d) L5

**3.** India became the 4th country to demonstrate space docking through mission:

(a) NISAR (b) Gaganyaan (c) SpaDeX (d) GSAT-20

**4.** India's first indigenous aircraft carrier is:

(a) INS Vikramaditya (b) INS Viraat (c) INS Vikrant (d) INS Arihant

**5.** Which ISRO mission was India's first interplanetary mission?

(a) Chandrayaan-1 (b) Chandrayaan-2 (c) Aditya-L1 (d) Mars Orbiter Mission (Mangalyaan)

**6.** Mission Divyastra (March 2024) demonstrated which capability of Agni-V?

(a) Hypersonic glide (b) MIRV (Multiple Independently Targetable Re-entry Vehicles) (c) Submarine launch (d) Anti-satellite

**7.** The Nobel Prize in Physics 2024 was awarded for work on:

(a) CRISPR gene editing (b) Foundational discoveries enabling machine learning with artificial neural networks (c) Quantum entanglement (d) Gravitational waves

**8.** What does "DNA" in ZyCoV-D (Zydus COVID vaccine) signify?

(a) It is made from inactivated virus (b) It is a DNA plasmid vaccine — the world's first DNA vaccine (c) It delivers mRNA (d) It is a protein subunit vaccine

**9.** India's net-zero emissions target year is:

(a) 2050 (b) 2060 (c) 2070 (d) 2075

**10.** Which layer of the OSI model is responsible for routing packets between networks?

(a) Data Link (b) Network (c) Transport (d) Session

**11.** 1 Terabyte (TB) equals how many Gigabytes?

(a) 1,000 (b) 1,024 (c) 1,048 (d) 10,000

**12.** AlphaFold, the AI tool that predicted protein structures, was developed by:

(a) OpenAI (b) Microsoft Research (c) Google DeepMind (d) IBM Watson

**13.** The first CRISPR-based therapy (Casgevy) approved by the US FDA in December 2023 is used to treat:

(a) Alzheimer's disease (b) Sickle cell disease and beta-thalassemia (c) HIV/AIDS (d) Type 1 diabetes

**14.** NISAR is a joint Earth observation satellite of ISRO and:

(a) ESA (b) JAXA (c) NASA (d) Roscosmos

**15.** BrahMos is a supersonic cruise missile jointly developed by India and:

(a) USA (b) France (c) Russia (d) Israel

**16.** India's first nuclear submarine (INS Arihant) belongs to which class?

(a) Akula (b) Arihant (c) Chakra (d) Scorpene

**17.** Which Indian satellite launch vehicle is used for heavier geosynchronous satellites?

(a) PSLV (b) SSLV (c) GSLV (d) SLV-3

**18.** India's National Space Day is celebrated on:

(a) 15 August (b) 22 October (c) 23 August (d) 14 November

**19.** The Bhadla Solar Park, the world's largest solar park, is located in:

(a) Gujarat (b) Tamil Nadu (c) Karnataka (d) Rajasthan

**20.** India's 3-stage nuclear programme was designed by:

(a) A.P.J. Abdul Kalam (b) Homi J. Bhabha (c) Vikram Sarabhai (d) Raja Ramanna

**21.** Which protocol is used for sending emails?

(a) HTTP (b) FTP (c) SMTP (d) POP3

**22.** The first generation of computers used:

(a) Transistors (b) Integrated circuits (c) Vacuum tubes (d) Microprocessors

**23.** World's first DNA vaccine for COVID-19 was developed by:

(a) Serum Institute (b) Bharat Biotech (c) Zydus Lifesciences (d) ICMR

**24.** India's target for Green Hydrogen production by 2030 is:

(a) 1 MMT/yr (b) 3 MMT/yr (c) 5 MMT/yr (d) 10 MMT/yr

**25.** The Nobel Peace Prize 2024 was awarded to:

(a) Greta Thunberg (b) WHO (c) Nihon Hidankyo (Japanese atomic bomb survivors' organisation) (d) Julian Assange

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### Answer Key

1-b | 2-a | 3-c | 4-c | 5-d | 6-b | 7-b | 8-b | 9-c | 10-b | 11-b | 12-c | 13-b | 14-c | 15-c | 16-b | 17-c | 18-c | 19-d | 20-b | 21-c | 22-c | 23-c | 24-c | 25-c

**Score:** 23–25 = Excellent · 18–22 = Good · Below 18 = Re-read Parts A (Space), B (Defence), and T (Computer Awareness).

**EXAM ALERT**

**Error-log rule:** For every question you got wrong, write one sentence explaining what you missed. Retest yourself on only those questions 3 days later — without looking at the solution. Re-reading alone does not fix the gap; active recall does.

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*Review the "What's New" section before every exam — ISRO and DRDO add new milestones regularly and Nobel laureates are announced each October.*

# PART T — COMPUTER AWARENESS DEEP-DRILL (Banks GA + SSC + RRB)

> *Computer Awareness carries 5-10 marks in Banks Mains GA, 2-3 in SSC CHSL/CGL Tier-2 (Computer Knowledge), 2-3 in RRB NTPC. Drill these tables to mastery.*

## T.1 Hardware components (revisit)

| Component | Function |
|---|---|
| CPU | Brain; CU + ALU + Registers |
| RAM | Volatile working memory; lost on power-off |
| ROM | Non-volatile boot storage; PROM, EPROM, EEPROM variants |
| Cache | Fast SRAM near CPU (L1, L2, L3) |
| HDD | Magnetic platter storage; mechanical |
| SSD | Flash-based; faster, no moving parts |
| GPU | Graphics + parallel computation (AI training) |
| Motherboard | Main circuit board |
| Power Supply (SMPS) | Converts AC to DC |
| Monitor / Display | Output (LCD, LED, OLED, AMOLED, CRT) |
| Keyboard, Mouse | Input devices |
| Scanner | Input (image / OCR) |
| Printer | Output (Inkjet, Laser, Dot Matrix, Plotter, 3-D) |
| Speaker | Output (audio) |
| Microphone | Input (audio) |
| Webcam | Input (video) |
| UPS | Uninterruptible Power Supply (battery backup) |

## T.2 Memory hierarchy + units (must know cold)

| Unit | Bytes | Notes |
|---|---|---|
| 1 nibble | 4 bits | half-byte |
| 1 byte | 8 bits | smallest addressable |
| 1 KB | 1,024 B | ($10^3$ in SI) |

| Unit | Bytes | Notes |
|---|---|---|
| 1 MB | 1,024 KB | |
| 1 GB | 1,024 MB | |
| 1 TB | 1,024 GB | |
| 1 PB | 1,024 TB | petabyte |
| 1 EB | 1,024 PB | exabyte |
| 1 ZB | 1,024 EB | zettabyte |
| 1 YB | 1,024 ZB | yottabyte |
| 1 word | 16, 32, or 64 bits | machine-dependent |

## T.3 Software taxonomy

| Type | Examples |
|---|---|
| **System software** | OS (Windows, Linux, macOS, Android, iOS); device drivers; utility (antivirus, defragmenter); language translators (assembler, compiler, interpreter, linker, loader) |
| **Application software** | MS Office, Adobe Photoshop, Tally, web browsers, email clients, games |
| **Open-source** | Linux, Apache, MySQL, Python, Firefox, LibreOffice |
| **Proprietary** | Windows, MS Office, Photoshop, Oracle DB |
| **Freeware** | Free to use; not necessarily open source (e.g., Skype, WhatsApp) |
| **Shareware** | Try-before-buy |
| **Firmware** | Embedded software in hardware (BIOS, router firmware) |
| **Middleware** | Connects different software (DBMS-app middleware) |

## T.4 Operating systems

- **Linux** — Linus Torvalds, 1991; Ubuntu, Debian, Fedora, Red Hat, CentOS, Arch.
- **Windows** — Microsoft (1985-): 1.0, 95, 98, XP, Vista, 7, 8, 10, 11.
- **macOS** — Apple (UNIX-based).
- **Android** — Google (Linux-based; mobile dominant).
- **iOS** — Apple (mobile).
- **Chrome OS** — Google (browser-based).
- **BharatOS (BHRAT-OS)** — IIT Madras + JandKops; Indian mobile OS (2023).

## T.5 MS Office shortcuts (high-yield)

### Universal

| Shortcut | Action |
|---|---|
| Ctrl+A | Select all |
| Ctrl+C | Copy |
| Ctrl+X | Cut |
| Ctrl+V | Paste |
| Ctrl+Z | Undo |
| Ctrl+Y | Redo |
| Ctrl+F | Find |
| Ctrl+H | Find + Replace |
| Ctrl+S | Save |
| Ctrl+P | Print |
| Ctrl+N | New |
| Ctrl+O | Open |
| Ctrl+W | Close window |
| Ctrl+B | Bold |
| Ctrl+I | Italic |
| Ctrl+U | Underline |
| Alt+Tab | Switch windows |
| Win+L | Lock screen |
| Win+D | Show desktop |
| F1 | Help |
| F5 | Refresh / Slideshow (PowerPoint) |
| F12 | Save As |

### Word

| Shortcut | Action |
|---|---|
| Ctrl+Shift+L | Apply bullets |
| Ctrl+Shift+N | Apply normal style |

| Shortcut | Action |
|---|---|
| Ctrl+E | Centre alignment |
| Ctrl+J | Justify |
| Ctrl+L | Left-align |
| Ctrl+R | Right-align |
| Ctrl+Enter | Page break |
| Shift+F3 | Toggle case |
| Ctrl+Shift+W | Underline words only |
| Ctrl+Shift+P | Change font size |

## Excel

| Shortcut | Action |
|---|---|
| Ctrl+; | Insert today's date |
| Ctrl+Shift+; | Insert current time |
| F4 | Toggle absolute reference |
| F2 | Edit cell |
| Ctrl+1 | Format Cells dialog |
| Alt+= | AutoSum |
| Ctrl+arrow | Jump to data edge |
| Ctrl+Shift+L | Toggle filter |
| Ctrl+T | Create table |
| F11 | Quick chart |
| Alt+Enter | New line in cell |

## PowerPoint

| Shortcut | Action |
|---|---|
| F5 | Start from beginning |
| Shift+F5 | Start from current slide |
| Esc | Exit slideshow |
| B / W | Black / white screen |
| Ctrl+M | New slide |

| Shortcut | Action |
|---|---|
| Ctrl+D | Duplicate slide |

## T.6 Networking — OSI 7 layers (memorise)

| Layer | Function | Protocol example |
|---|---|---|
| 7. **Application** | end-user services | HTTP, HTTPS, FTP, SMTP, DNS, SSH, POP3, IMAP, Telnet |
| 6. **Presentation** | encoding, encryption | SSL/TLS, JPEG, MPEG, ASCII, EBCDIC |
| 5. **Session** | session management | NetBIOS, RPC, SQL |
| 4. **Transport** | end-to-end reliability | TCP, UDP, SCTP |
| 3. **Network** | logical addressing, routing | IP, ICMP, OSPF, BGP, IGMP |
| 2. **Data Link** | physical addressing, frames | Ethernet, PPP, ARP, MAC |
| 1. **Physical** | bits over medium | Cables, hubs, RJ-45, fibre |

> ***Mnemonic top→bottom**: "**A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing". **Mnemonic bottom→top**: "**P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way" (PDNTSPA).*

## T.7 TCP/IP (4 layers)

| TCP/IP | OSI mapping |
|---|---|
| Application | 7 + 6 + 5 |
| Transport | 4 |
| Internet | 3 |
| Network access | 2 + 1 |

## T.8 Network terms + ports

| Term | Meaning |
|---|---|
| LAN | Local Area Network (within building) |
| WAN | Wide Area Network (continents — internet) |
| MAN | Metropolitan Area Network (city) |
| PAN | Personal Area Network (Bluetooth) |
| WLAN | Wireless LAN (Wi-Fi) |
| VPN | Virtual Private Network (encrypted tunnel) |
| VLAN | Virtual LAN (logical segmentation) |

| Term | Meaning |
|---|---|
| Internet | global TCP/IP network |
| Intranet | private network within an org |
| Extranet | intranet with external partners |
| World Wide Web (WWW) | hypertext document system on internet |

## T.9 Common port numbers (must memorise)

| Service | Port |
|---|---|
| HTTP | 80 |
| HTTPS | 443 |
| FTP | 20 (data), 21 (control) |
| SSH | 22 |
| Telnet | 23 |
| SMTP (mail send) | 25 |
| DNS | 53 |
| DHCP | 67/68 |
| HTTP-Proxy | 8080 |
| POP3 (mail receive) | 110 |
| IMAP | 143 |
| SQL Server | 1433 |
| MySQL | 3306 |
| PostgreSQL | 5432 |
| MongoDB | 27017 |
| RDP (Remote Desktop) | 3389 |

## T.10 IP addressing

- **IPv4** — 32-bit (4 octets); ~4.3 billion addresses; format e.g., `192.168.1.1`.
- **IPv6** — 128-bit; $3.4 \times 10^{38}$ addresses; format e.g., `2001:0db8:85a3::8a2e:0370:7334`.
- **Private IP ranges** (IPv4): `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`.
- **Loopback**: `127.0.0.1` (localhost).
- **MAC address** — 48-bit hardware address (e.g., `00:1A:2B:3C:4D:5E`).
- **DNS** — Domain Name System (resolves names to IPs).

- **DHCP** — Dynamic Host Configuration Protocol (auto IP assignment).
- **NAT** — Network Address Translation (private → public IP).

## T.11 Internet milestones (memorise)

| Year | Event |
|---|---|
| 1969 | ARPANET (DARPA, USA) — birth of internet |
| 1971 | First email (Ray Tomlinson); `@` chosen |
| 1983 | TCP/IP becomes standard |
| **1989** | **WWW proposed** (Tim Berners-Lee at CERN) |
| 1990 | First web browser ("WorldWideWeb" by Berners-Lee) |
| 1993 | Mosaic browser |
| 1995 | India's first internet (VSNL) on **15 Aug 1995** |
| 1998 | Google founded (Larry Page + Sergey Brin) |
| 2004 | Facebook founded (Mark Zuckerberg) |
| 2005 | YouTube founded |
| 2006 | Twitter founded |
| 2007 | iPhone launched |
| 2008 | Android released |
| 2010 | Instagram founded |
| 2017 | Reliance Jio rolls out 4G in India |
| 2022 | 5G commercial launch in India (Oct) |
| 2022 | ChatGPT launched (Nov 30, OpenAI) |
| 2024 | India crosses **1 billion internet users**; UPI cross-border live in 7 countries |

## T.12 Programming languages

| Language | Year | Creator | Use |
|---|---|---|---|
| FORTRAN | 1957 | John Backus (IBM) | Scientific computing |
| LISP | 1958 | John McCarthy | AI |
| COBOL | 1959 | Grace Hopper | Business |
| BASIC | 1964 | Kemeny + Kurtz | Beginners |
| C | 1972 | Dennis Ritchie | System programming, Unix |

| Language | Year | Creator | Use |
|---|---|---|---|
| C++ | 1985 | Bjarne Stroustrup | Object-oriented |
| Java | 1995 | James Gosling (Sun) | Cross-platform; "Write once, run anywhere" |
| JavaScript | 1995 | Brendan Eich (Netscape) | Web browser |
| Python | 1991 | Guido van Rossum | General purpose, AI/ML |
| PHP | 1994 | Rasmus Lerdorf | Web backend |
| Ruby | 1995 | Yukihiro Matsumoto | Web (Ruby on Rails) |
| Swift | 2014 | Apple | iOS apps |
| Kotlin | 2011 | JetBrains | Android |
| Go (Golang) | 2009 | Google | Backend systems |
| Rust | 2010 | Mozilla | Systems, safe concurrency |
| TypeScript | 2012 | Microsoft | JavaScript with types |
| R | 1993 | Ihaka + Gentleman | Statistics, data science |
| MATLAB | 1984 | Cleve Moler | Numerical computing |
| SQL | 1974 | IBM (Codd's relational model) | Database queries |
| HTML | 1993 | Tim Berners-Lee | Web markup |

## T.13 Database concepts

- **DBMS** — Database Management System (Oracle, MySQL, MS SQL, PostgreSQL).
- **RDBMS** — Relational DBMS; based on E.F. Codd's relational model (1970).
- **NoSQL** — non-relational (MongoDB, Cassandra, Redis).
- **SQL commands**: SELECT, INSERT, UPDATE, DELETE, CREATE, DROP, ALTER, JOIN.
- **Primary key** — unique identifier for a row.
- **Foreign key** — references another table's primary key.
- **Normalisation** — eliminating redundancy (1NF, 2NF, 3NF, BCNF).
- **ACID** — Atomicity, Consistency, Isolation, Durability (transaction properties).

## T.14 Cybersecurity terms

| Term | Meaning |
|---|---|
| **Virus** | Self-replicating malicious code attached to host file |
| **Worm** | Standalone self-replicating program |
| **Trojan** | Disguised malware |
| **Ransomware** | Encrypts files; demands ransom (e.g., WannaCry 2017) |

| Term | Meaning |
|---|---|
| **Spyware** | Secretly monitors user |
| **Adware** | Shows unwanted ads |
| **Phishing** | Fake emails to steal credentials |
| **Spear phishing** | Targeted phishing |
| **Smishing** | Phishing via SMS |
| **Vishing** | Phishing via voice call |
| **DDoS** | Distributed Denial of Service (overwhelm server) |
| **Man-in-the-middle** | Intercepting communication |
| **SQL injection** | Inserting malicious SQL into input |
| **XSS** | Cross-Site Scripting |
| **Zero-day** | Newly discovered vulnerability before patch |
| **Firewall** | Filters network traffic |
| **VPN** | Virtual Private Network (encrypted tunnel) |
| **Encryption** | Converting data to unreadable form (AES, RSA) |
| **Symmetric encryption** | Same key for encrypt+decrypt (AES) |
| **Asymmetric encryption** | Public+private key pair (RSA) |
| **Hash** | One-way function (MD5, SHA-256) |
| **Digital signature** | Verifies authenticity + integrity |
| **2FA / MFA** | Two/Multi-factor authentication |
| **CAPTCHA** | "Completely Automated Public Turing test" |
| **IT Act 2000** | India's primary cyber law |
| **CERT-In** | Indian Computer Emergency Response Team |
| **Cyber Crime Reporting** | cybercrime.gov.in |

## T.15 Famous computer scientists

| Person | Contribution |
|---|---|
| **Charles Babbage** | "Father of computer"; Analytical Engine (1837) |
| **Ada Lovelace** | "First computer programmer" (Babbage's engine) |
| **Alan Turing** | Turing Machine (1936); cracked Enigma; Father of AI/CS |

| Person | Contribution |
|---|---|
| **John von Neumann** | Von Neumann architecture (stored-program) |
| **Grace Hopper** | COBOL; coined "bug"; Navy Rear Admiral |
| **Donald Knuth** | "Art of Computer Programming"; TeX |
| **Dennis Ritchie + Ken Thompson** | C language + Unix (Bell Labs); Turing Award |
| **Tim Berners-Lee** | World Wide Web (1989) |
| **Vint Cerf + Bob Kahn** | TCP/IP — "Fathers of internet" |
| **Linus Torvalds** | Linux kernel (1991) |
| **Bill Gates + Paul Allen** | Microsoft (1975); Windows |
| **Steve Jobs + Steve Wozniak** | Apple (1976); iPhone (2007) |
| **Mark Zuckerberg** | Facebook (2004) |
| **Larry Page + Sergey Brin** | Google (1998) |
| **Jeff Bezos** | Amazon (1994) |
| **Elon Musk** | PayPal, Tesla, SpaceX, X (Twitter), xAI |
| **Sundar Pichai** | Google CEO (2015-) |
| **Satya Nadella** | Microsoft CEO (2014-) |
| **Sam Altman** | OpenAI CEO; ChatGPT |
| **Geoffrey Hinton + Yann LeCun + Yoshua Bengio** | Deep learning pioneers; Turing Award 2018 |
| **N.R. Narayana Murthy** | Infosys co-founder |
| **Azim Premji** | Wipro Chairman emeritus |

## T.16 Computer Awareness mini-mock (15 Q)

1. Father of Computer? — Charles Babbage
2. WWW invented by? — Tim Berners-Lee (1989, CERN)
3. India's first internet day? — 15 Aug 1995 (VSNL)
4. Default port for HTTPS? — 443
5. RAM stands for? — Random Access Memory
6. ROM stands for? — Read Only Memory
7. Linux created by? — Linus Torvalds (1991)
8. Java created by? — James Gosling (Sun, 1995)
9. Python created by? — Guido van Rossum (1991)
10. India's IT Act year? — 2000

11. CERT-In stands for? — Indian Computer Emergency Response Team
12. ACID full form? — Atomicity, Consistency, Isolation, Durability
13. OSI total layers? — 7
14. SMTP port? — 25
15. India's mobile OS? — BharatOS (BHRAT-OS, IIT Madras + JandKops)

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*Computer Awareness covers ~5-10 marks in Banks Mains GA. These tables are mastery-level for that section.*